

# PRP

PRIVATE RUNTIME PLATFORM

## Architecture & Flow Atlas

v0.3.0 | Python-first / Reuse-before-build

34 diagram views spanning product boundaries, APIs, chat, speech, data, security, operations and delivery planning.

### REVISED DIRECTION

Compose qualified libraries and runtimes; write only proven gaps.

### PHASE 1

Chat and clip-based speech (ASR / approved preset TTS).

### PHASE 2

Image and video - separately gated extensions.

### INDEPENDENCE

Zuri, FUNG and Lalin Studio are optional clients, not PRP dependencies.

### EVIDENCE STATUS

DRAFT FOR REVIEW. Runtime experiments and qualification: NOT\_RUN.

### FRAMEWORK CHOICE

Candidate A/B evaluation is pending; no production framework is selected.

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Select a row to open a diagram

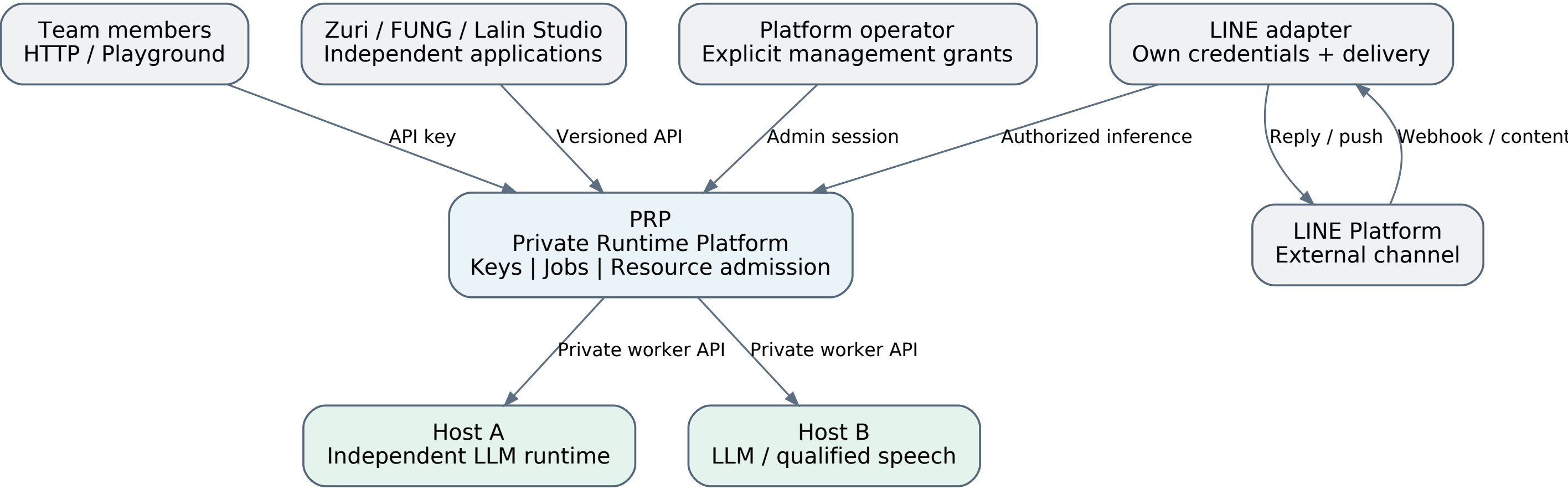
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Diagram labels are English for portability. Thai rationale, requirement links and editable SVG/DOT/sequence sources are in PRP-Diagram-Atlas.html and docs/DIAGRAMS-PRP.md.

D01 | System context

C4-inspired context

PRP-FR-001,002,054

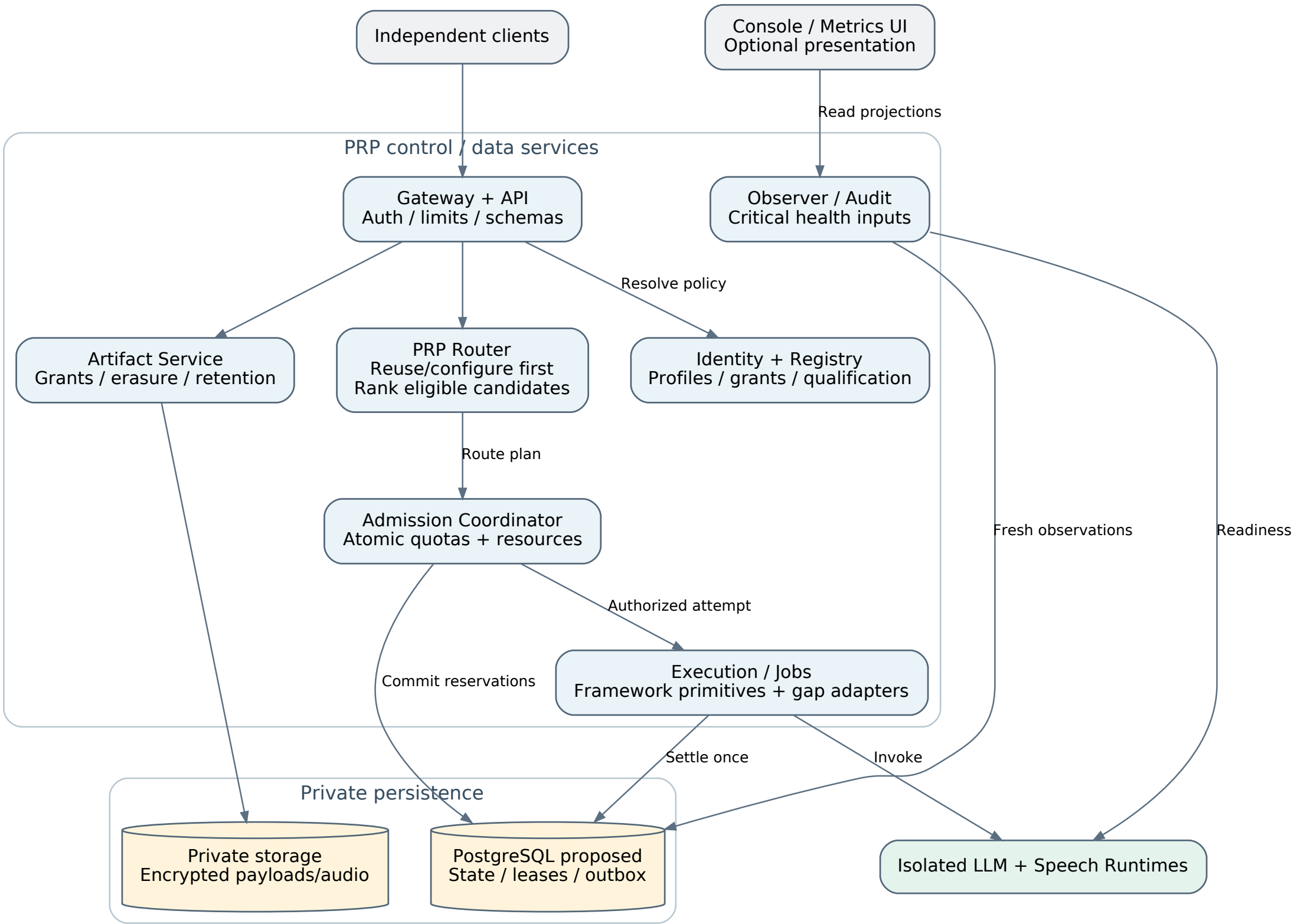


No mandatory business application. LINE traffic is outside the private compute boundary.

D02 | Containers and control/data planes

C4-inspired containers

PRP-FR-017,042,045,050

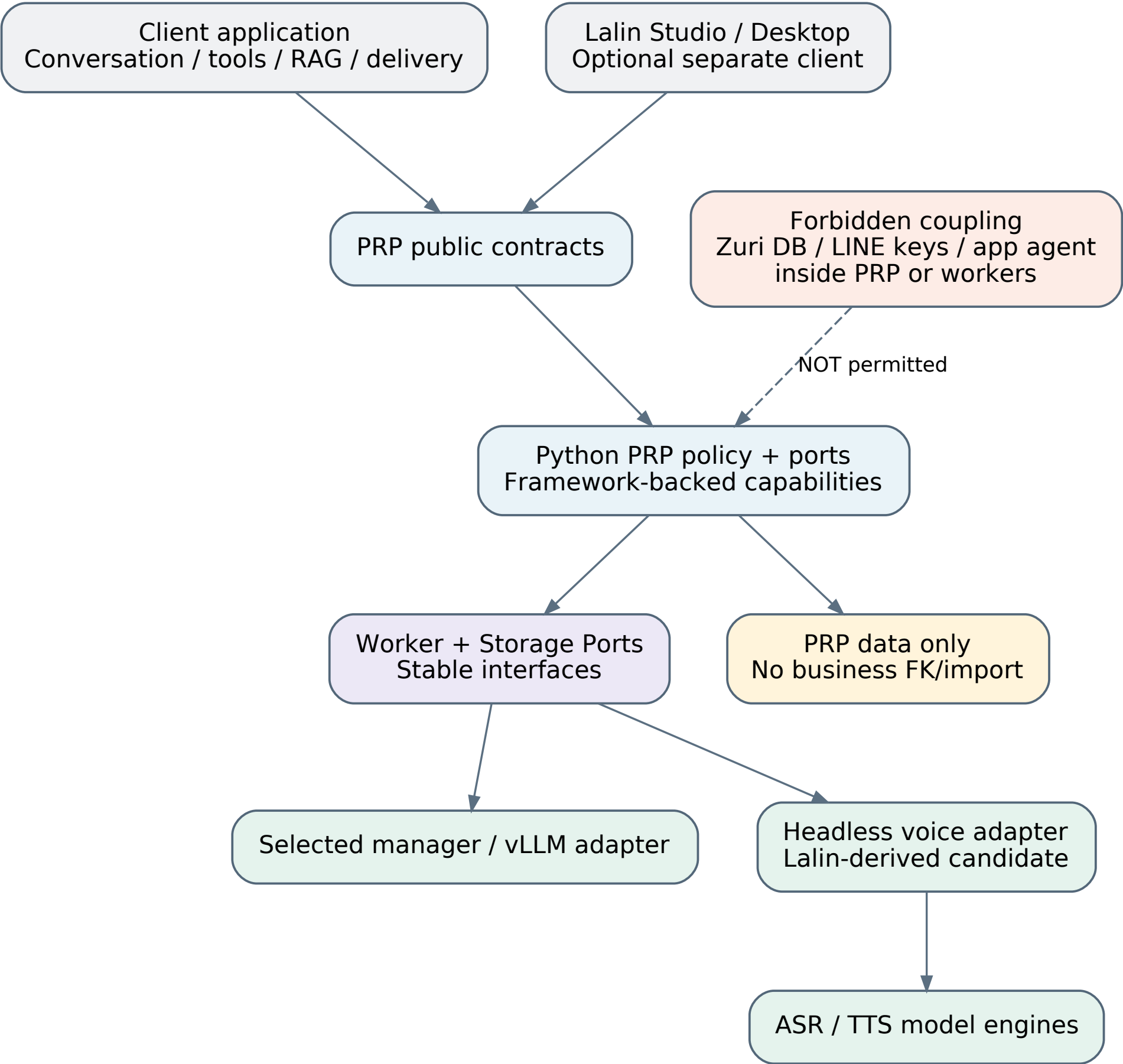


v0.3.0: logical ownership can be implemented by reused libraries/configuration; no automatic custom service.

D03 | Ownership and dependency rules

Component / package boundaries

PRP-FR-002,026,028,043,054; PRP-SEC-012

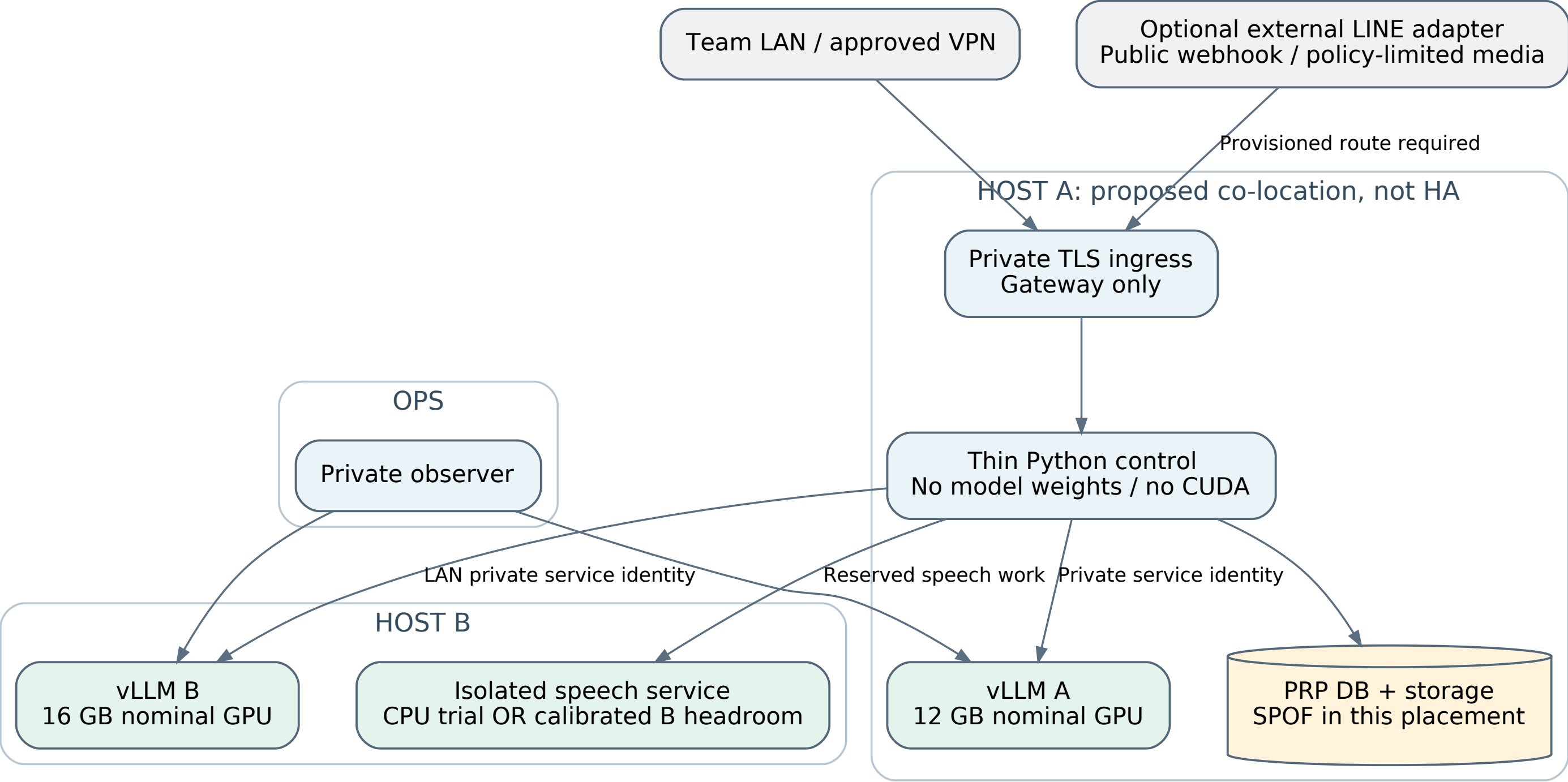


v0.3.0: logical ownership can be implemented by reused libraries/configuration; no automatic custom service.

D04 | Two-host private deployment

Deployment / network topology

PRP-FR-018,044,050; PRP-NFR-012

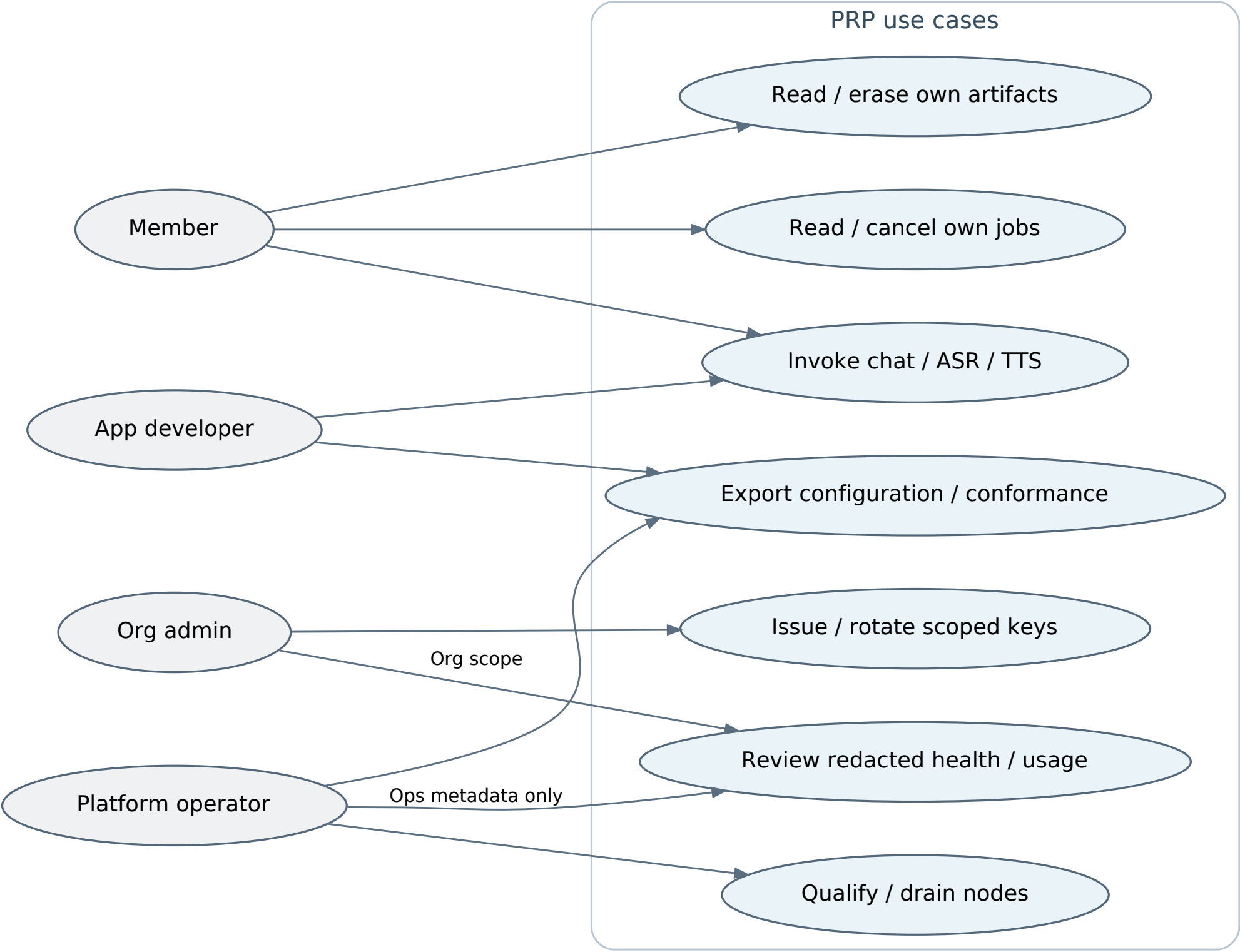


v0.3.0: logical ownership can be implemented by reused libraries/configuration; no automatic custom service.

# D05 | Actors and use cases

## Use-case view

PRP-FR-004,005,007,048

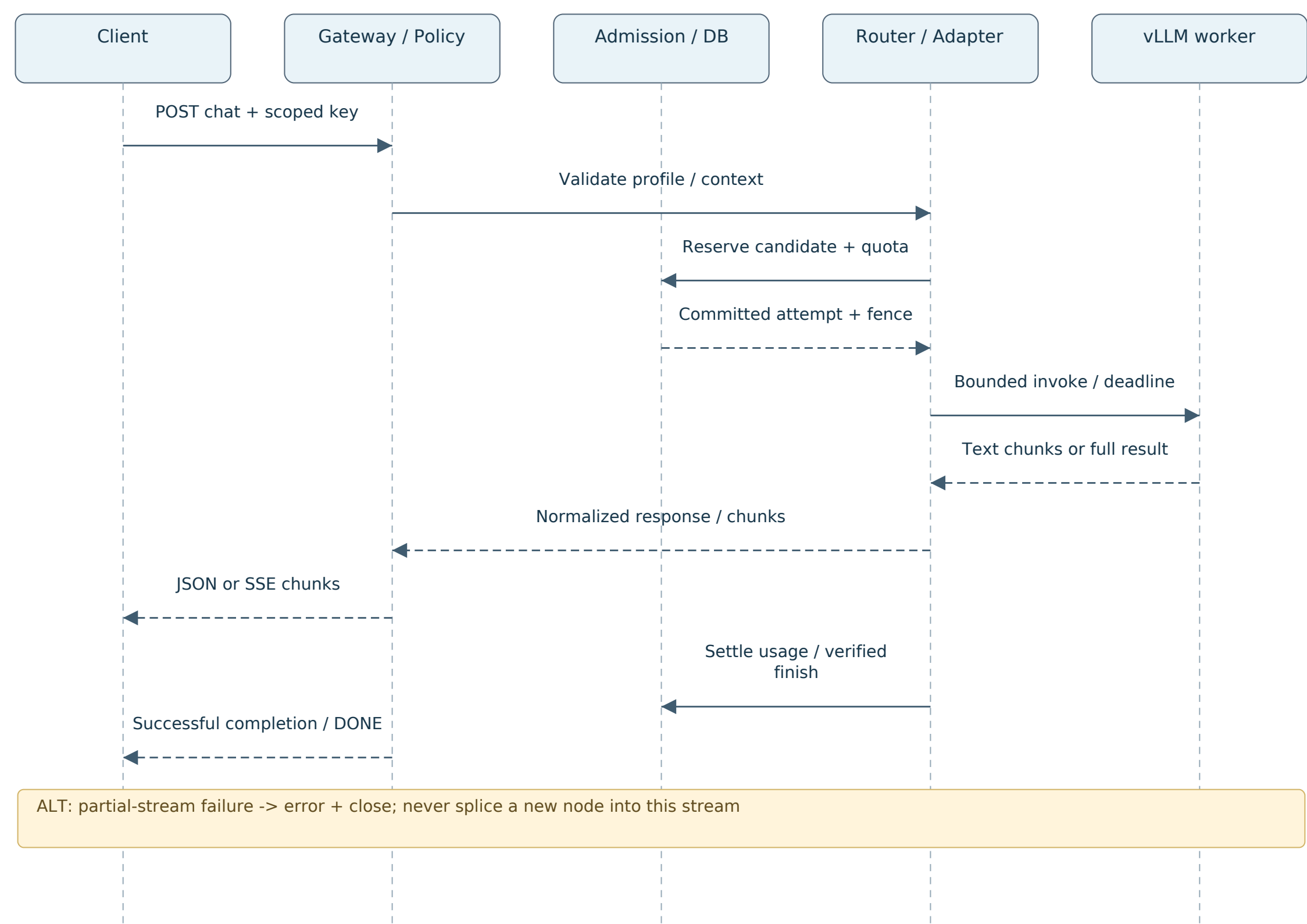


Target design. Framework bindings and hardware capacity require qualification before activation.

D06 | Chat JSON/SSE request

UML-style sequence

PRP-FR-017,024,025,027

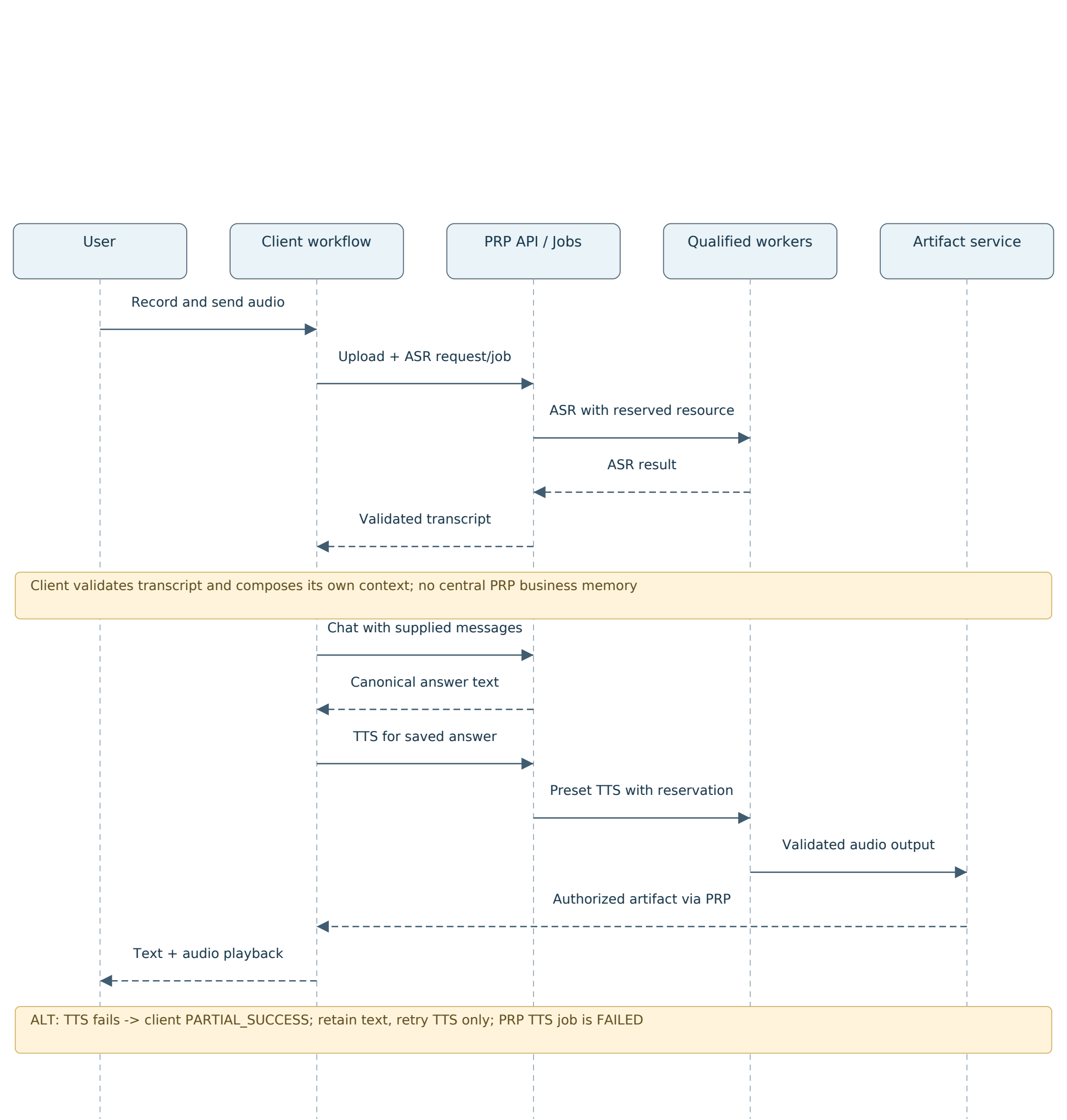


Target design. Framework bindings and hardware capacity require qualification before activation.

# D07 | Client-owned voice conversation

UML-style sequence

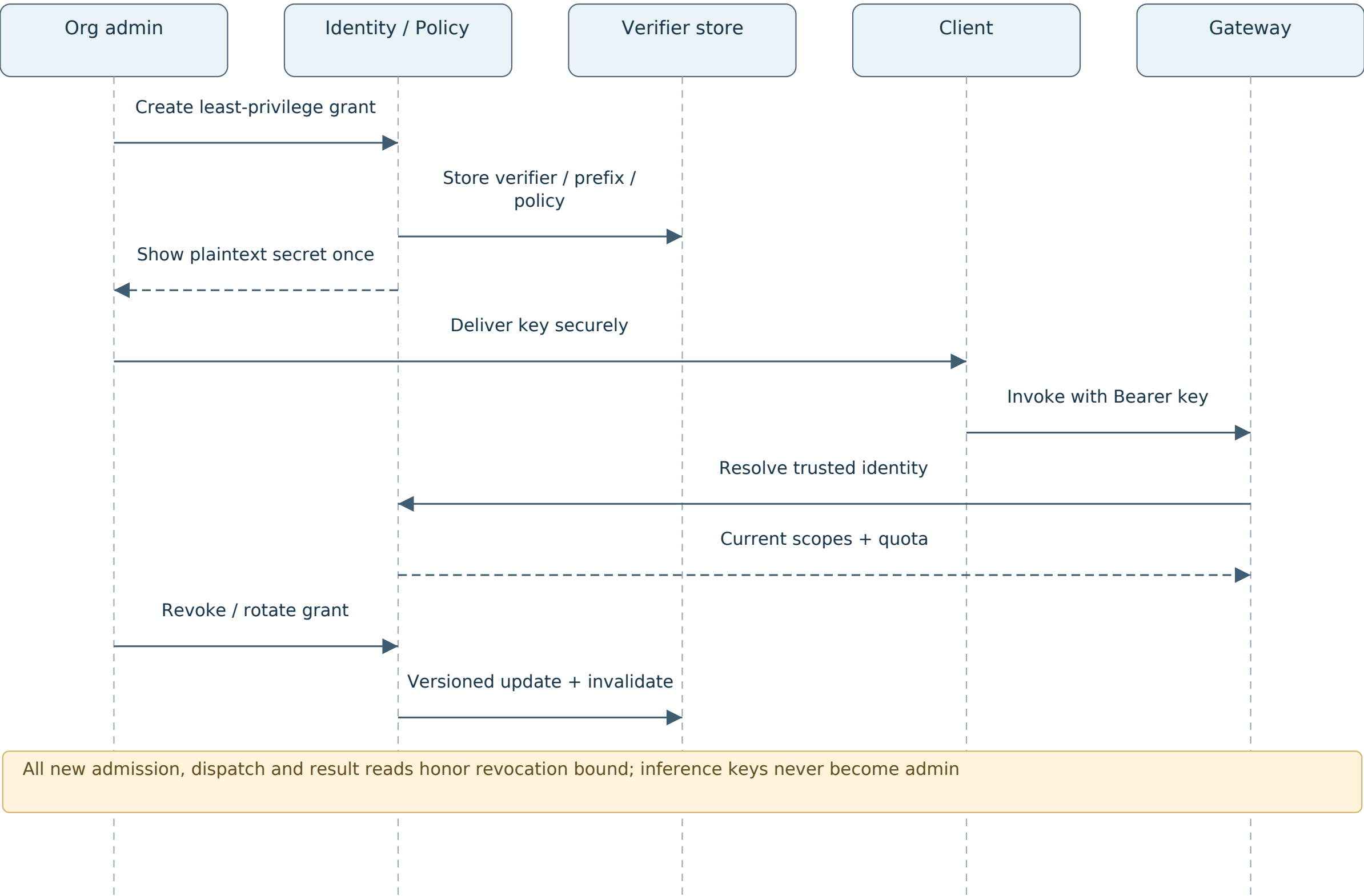
PRP-FR-031..038,049



# D08 | Key issuance, use and revocation

UML-style sequence

PRP-FR-005,006; PRP-SEC-002



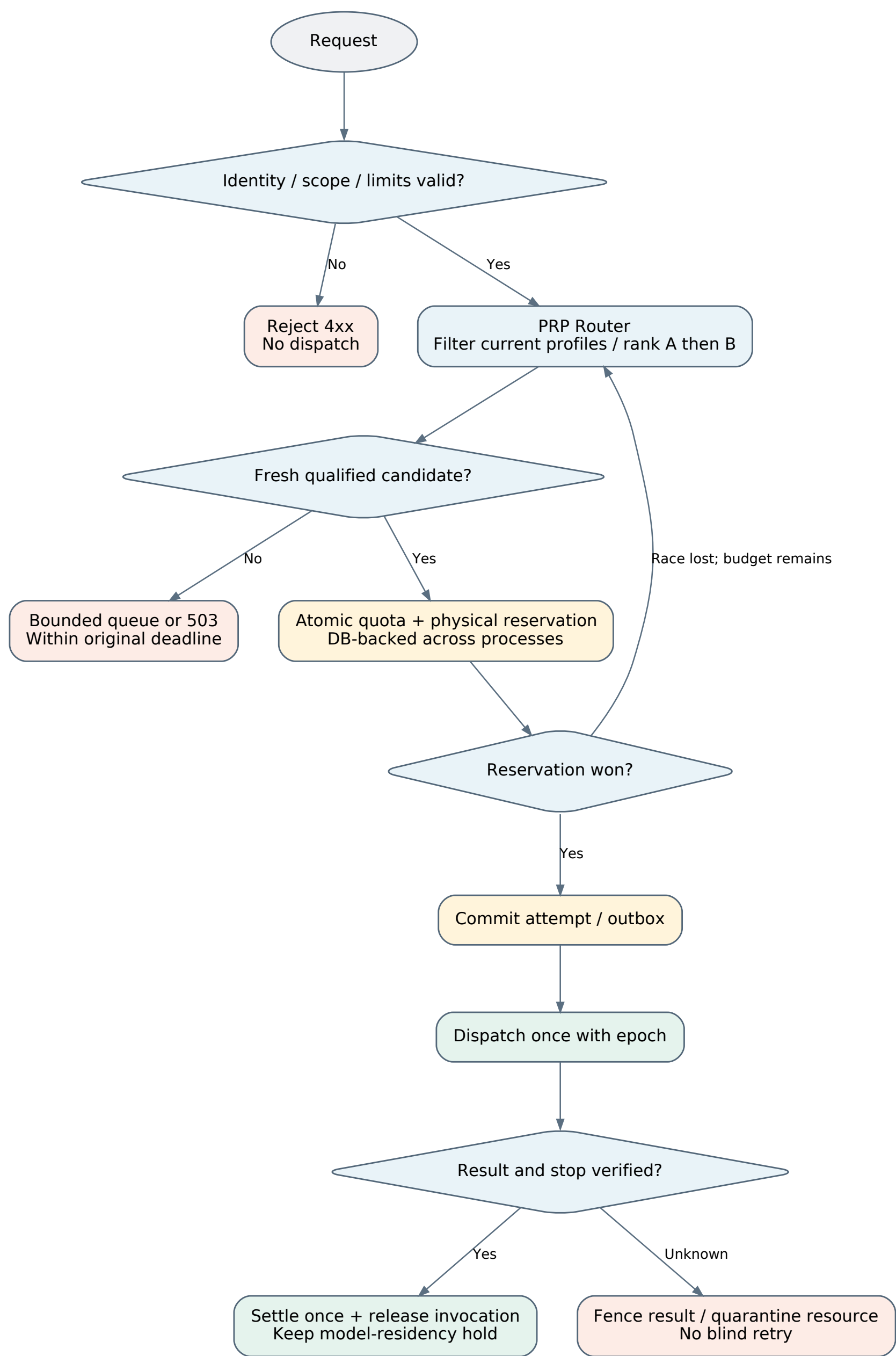
All new admission, dispatch and result reads honor revocation bound; inference keys never become admin

Target design. Framework bindings and hardware capacity require qualification before activation.

D09 | Admission and routing activity

Activity / decision flow

PRP-FR-008,016..022

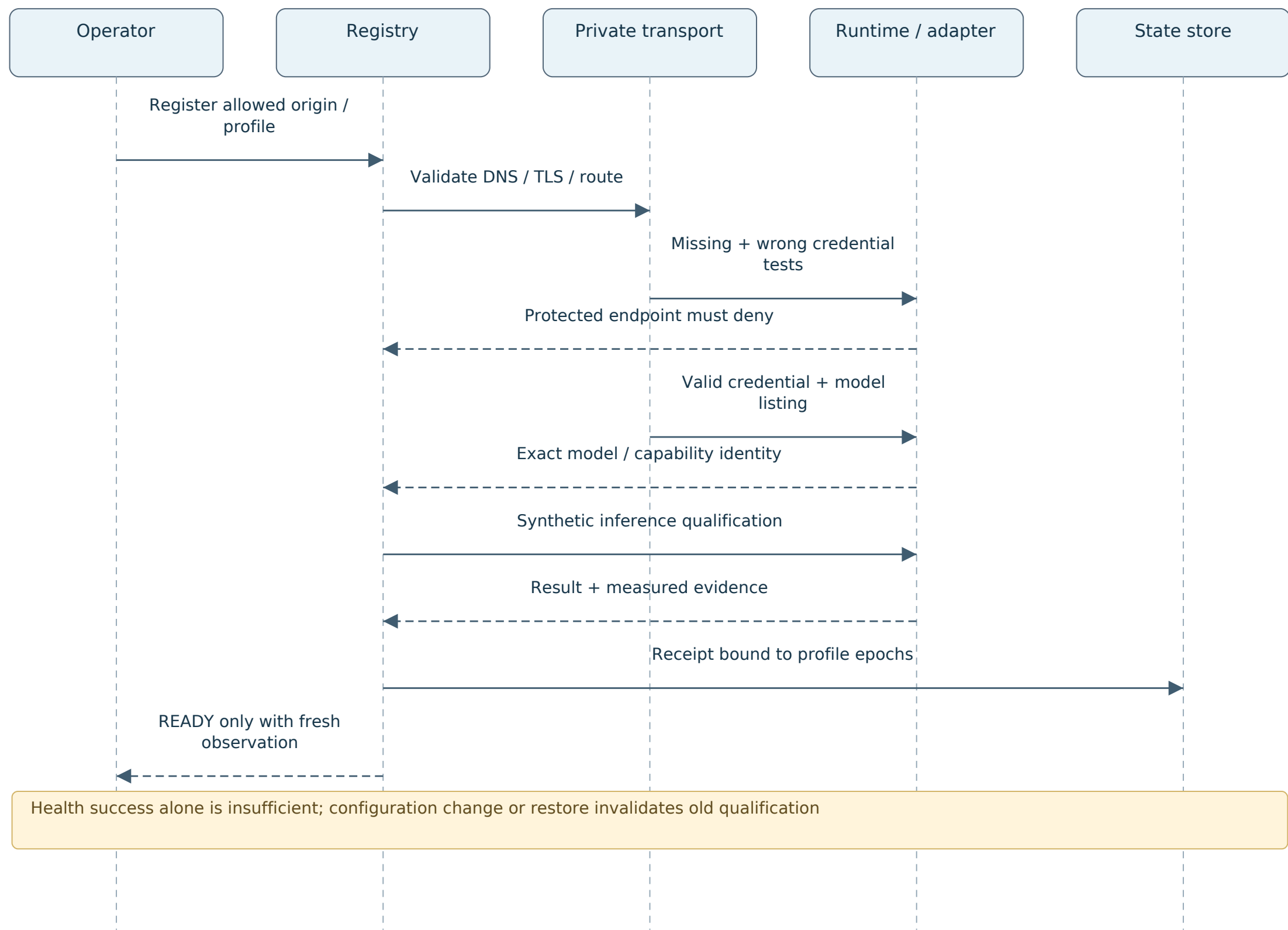


Target design. Framework bindings and hardware capacity require qualification before activation.

# D10 | Node enrollment and qualification

UML-style sequence

PRP-FR-010..015,042

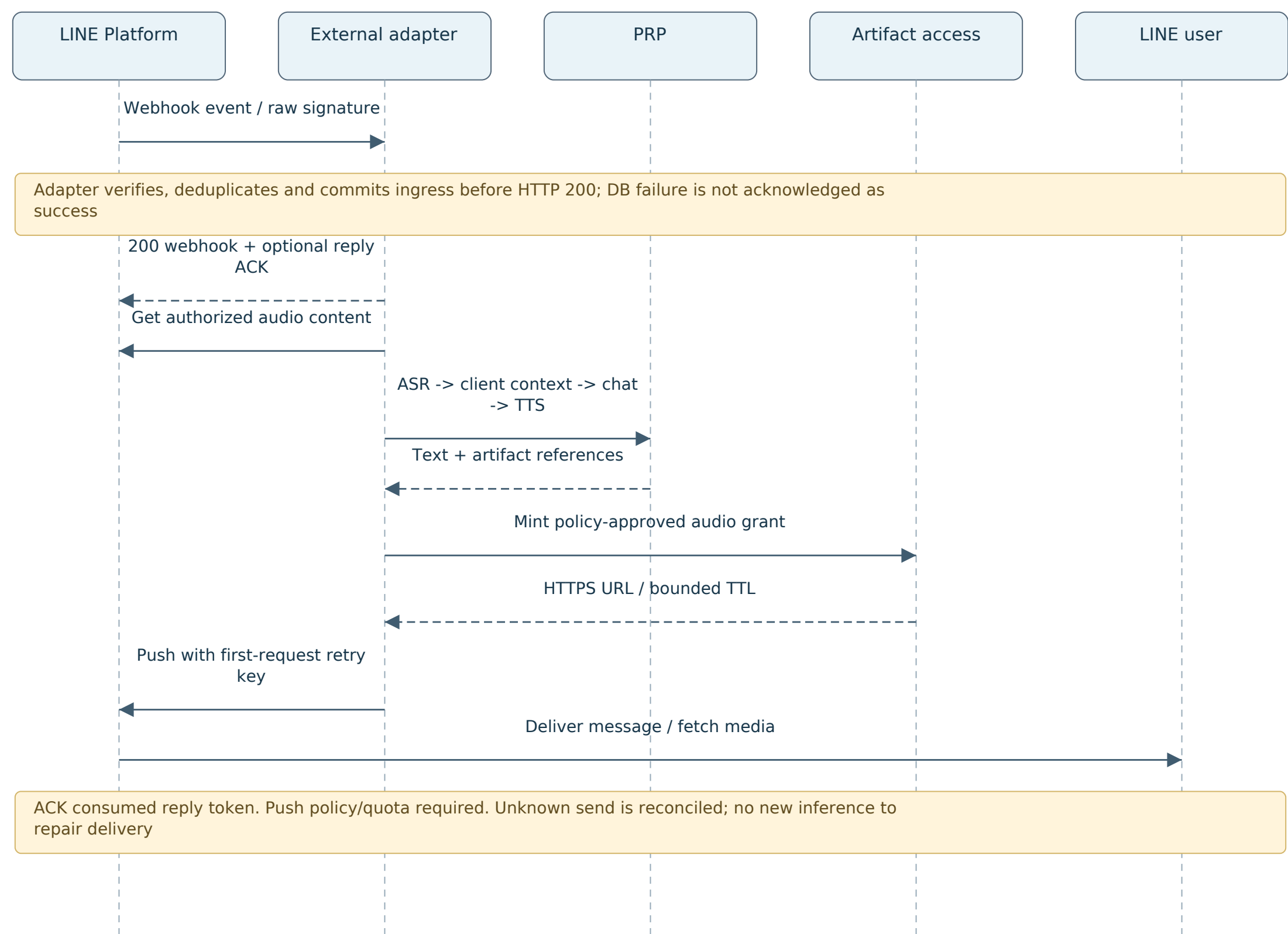


Target design. Framework bindings and hardware capacity require qualification before activation.

# D11 | LINE voice integration outside PRP

UML-style sequence

PRP-FR-040,055

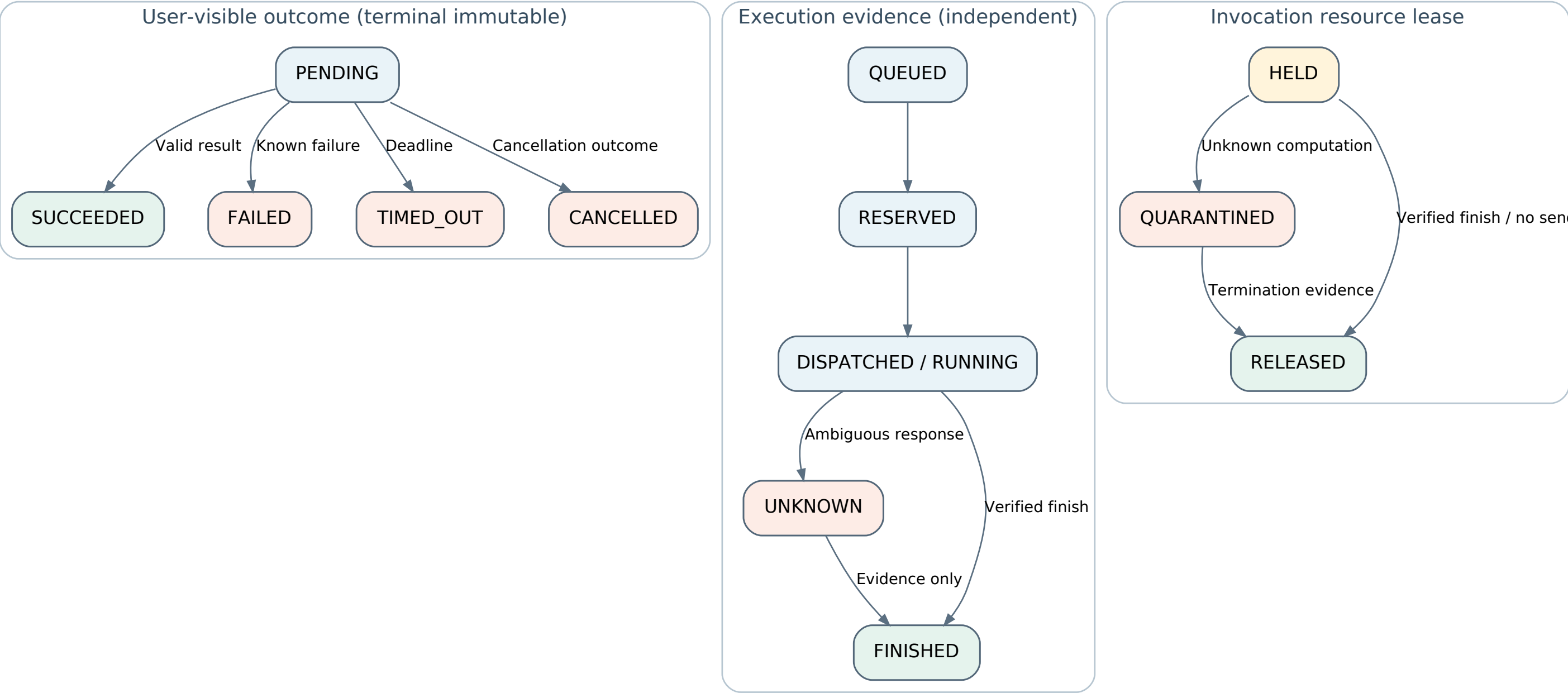


Target design. Framework bindings and hardware capacity require qualification before activation.

# D12 | Job state dimensions

State machines / product state

PRP-FR-020,021,036..038

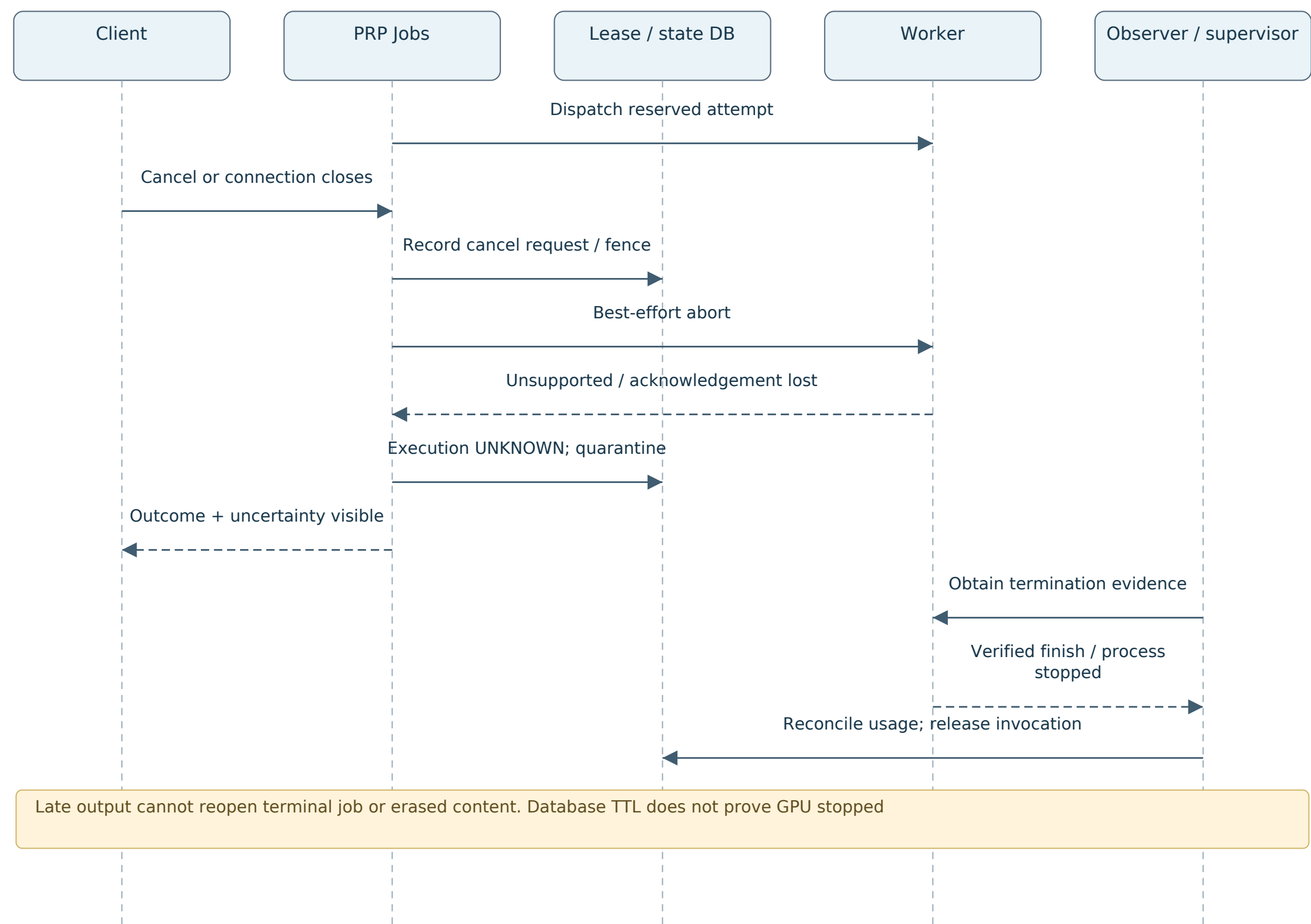


Target design. Framework bindings and hardware capacity require qualification before activation.

# D13 | Cancellation and uncertain recovery

UML-style sequence

PRP-FR-020..022,041

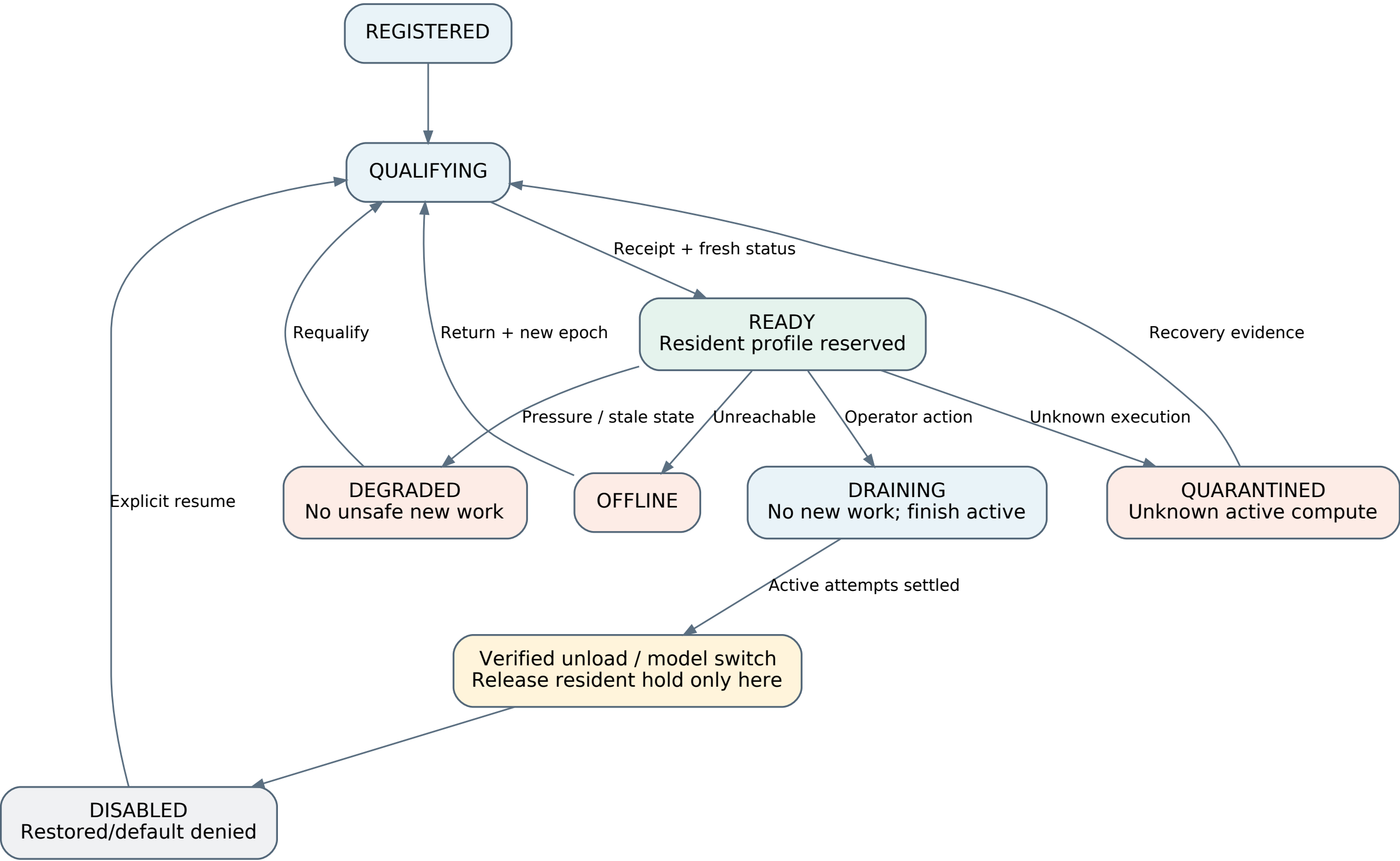


Target design. Framework bindings and hardware capacity require qualification before activation.

D14 | Runtime readiness and residency

State machine

PRP-FR-014,015,044

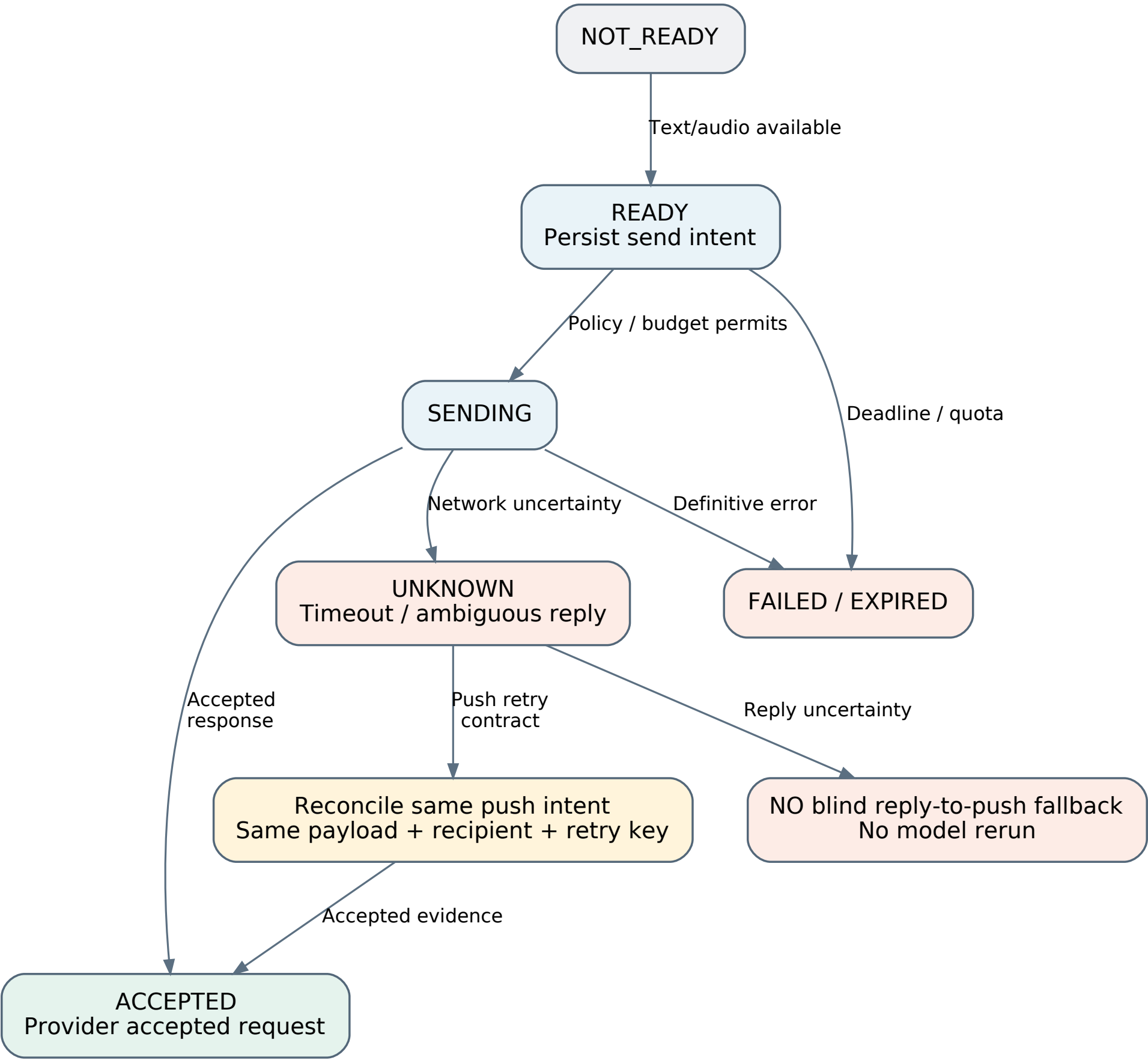


Target design. Framework bindings and hardware capacity require qualification before activation.

# D15 | External delivery lifecycle

State machine / external integration

PRP-FR-055

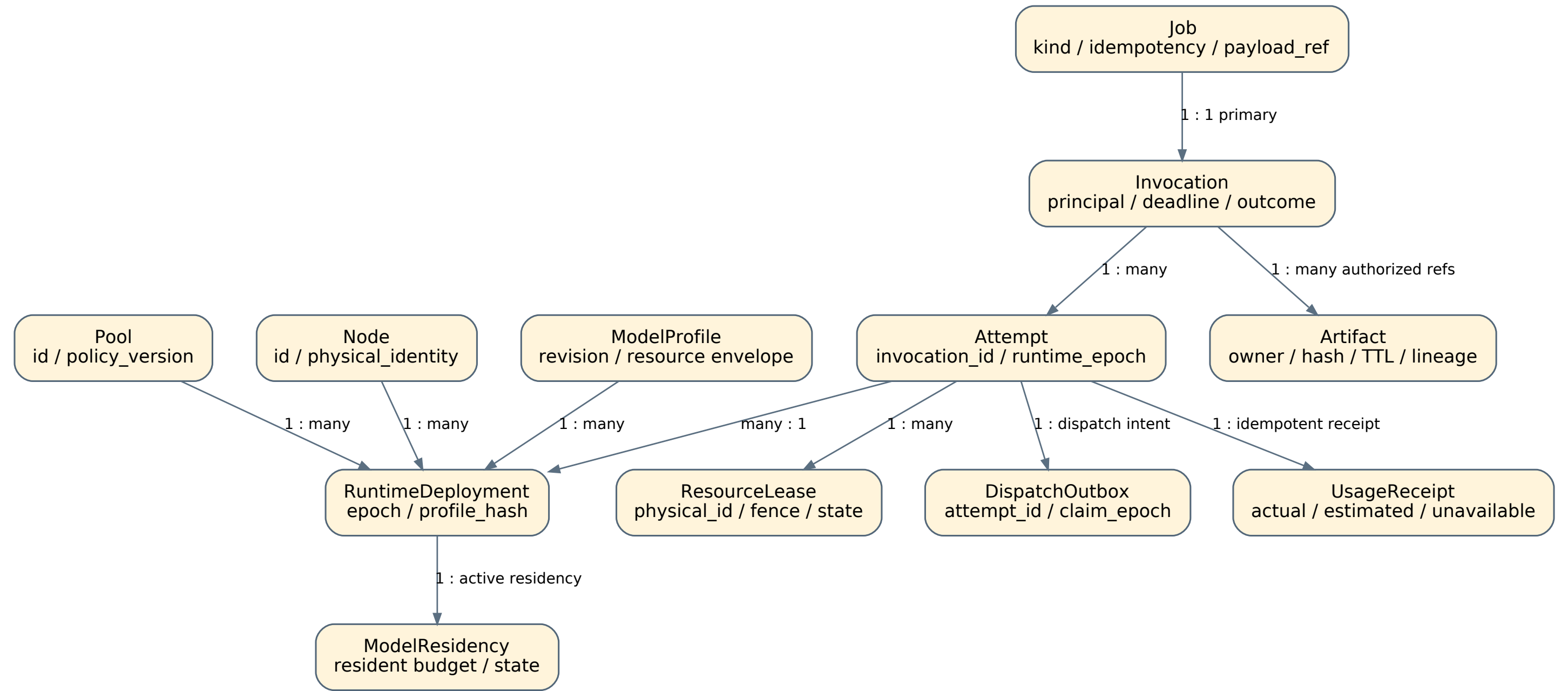


Target design. Framework bindings and hardware capacity require qualification before activation.

D16 | Compute and job data model

Logical ERD

PRP-FR-013,017,027,036..041

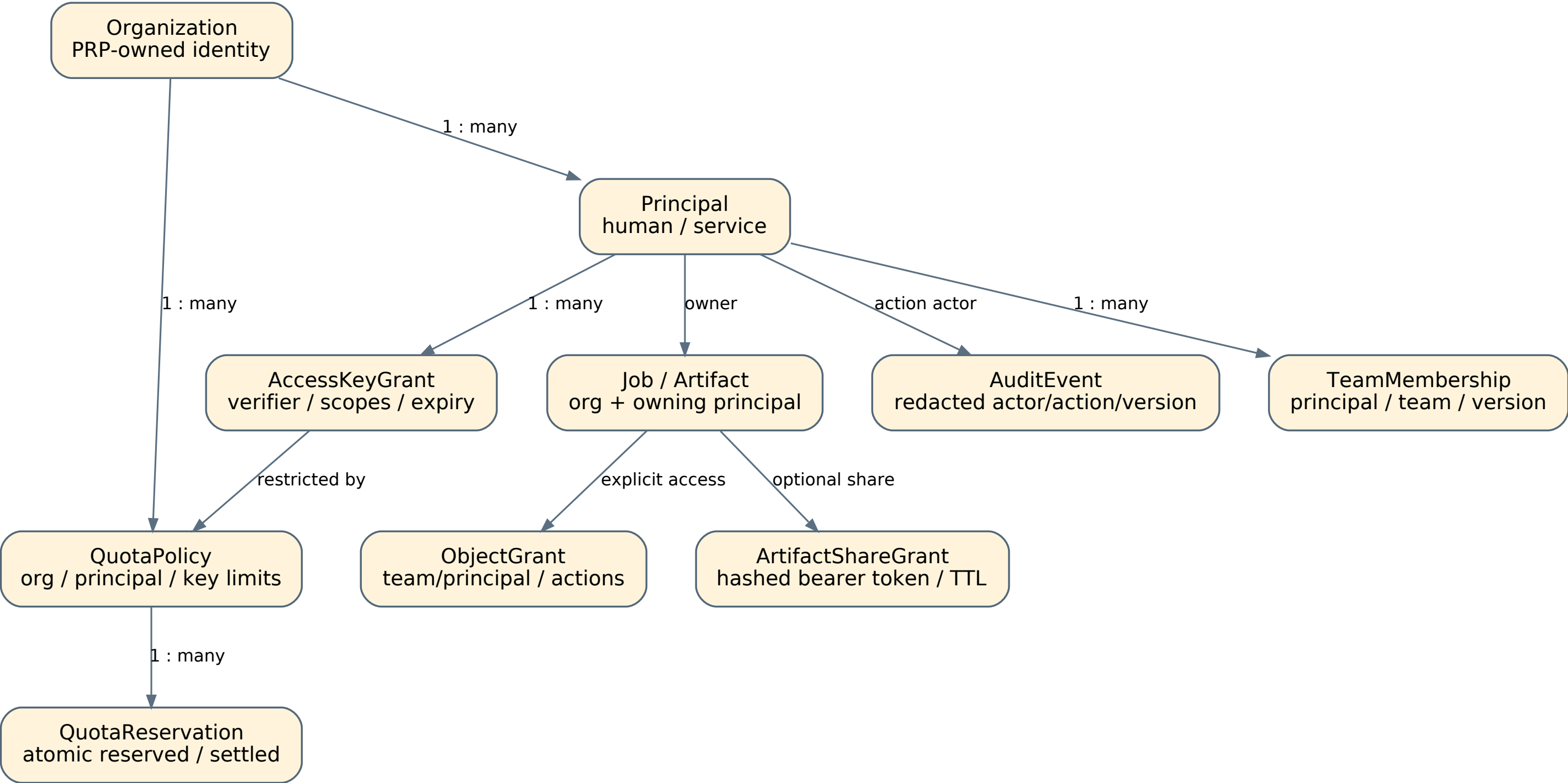


Target design. Framework bindings and hardware capacity require qualification before activation.

# D17 | Identity, keys and object grants

Logical ERD / authorization

PRP-FR-003..009,038..040; PRP-SEC-004

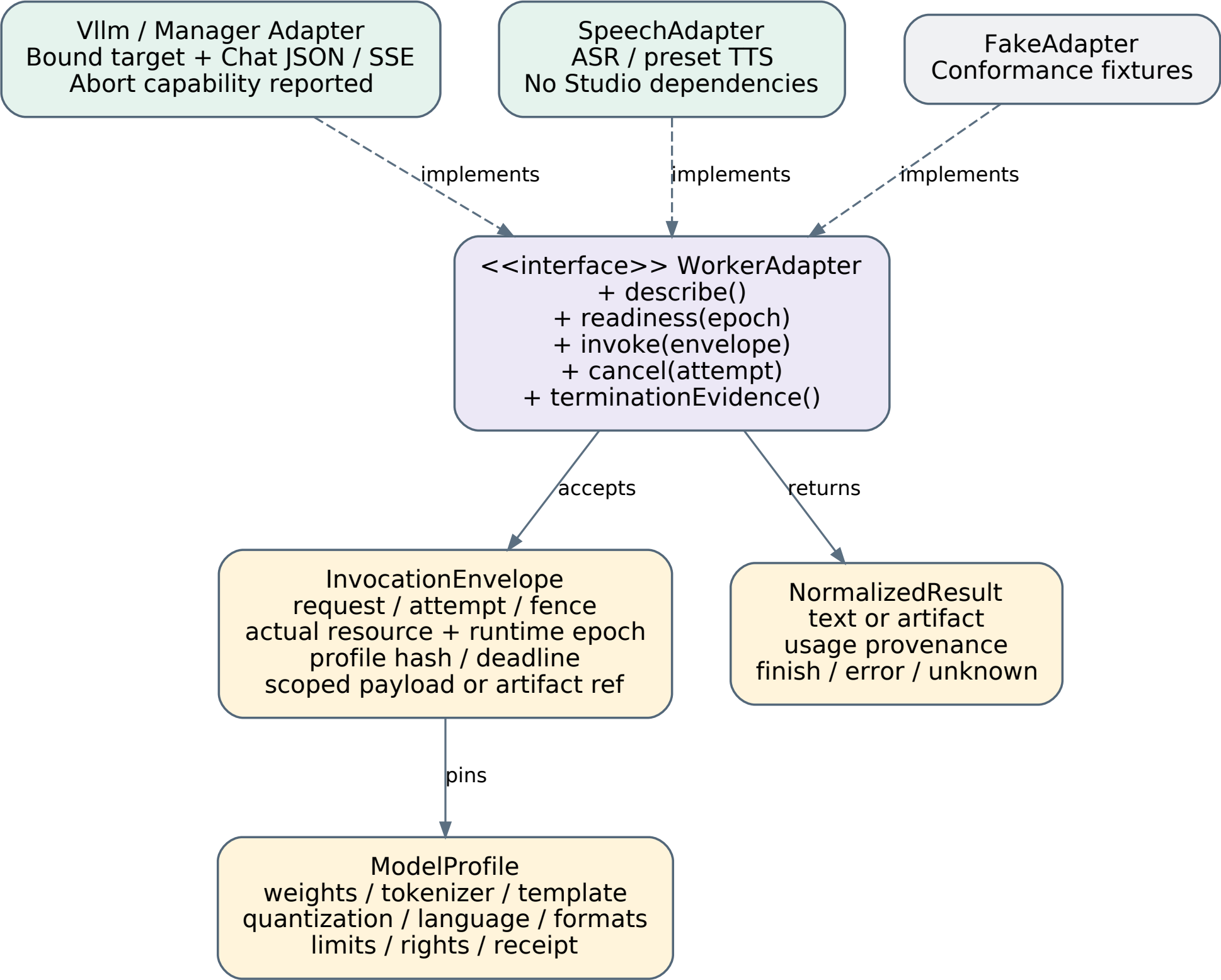


Target design. Framework bindings and hardware capacity require qualification before activation.

# D18 | Worker ports and profile contract

UML-inspired class/interface

PRP-FR-011,012,042,043

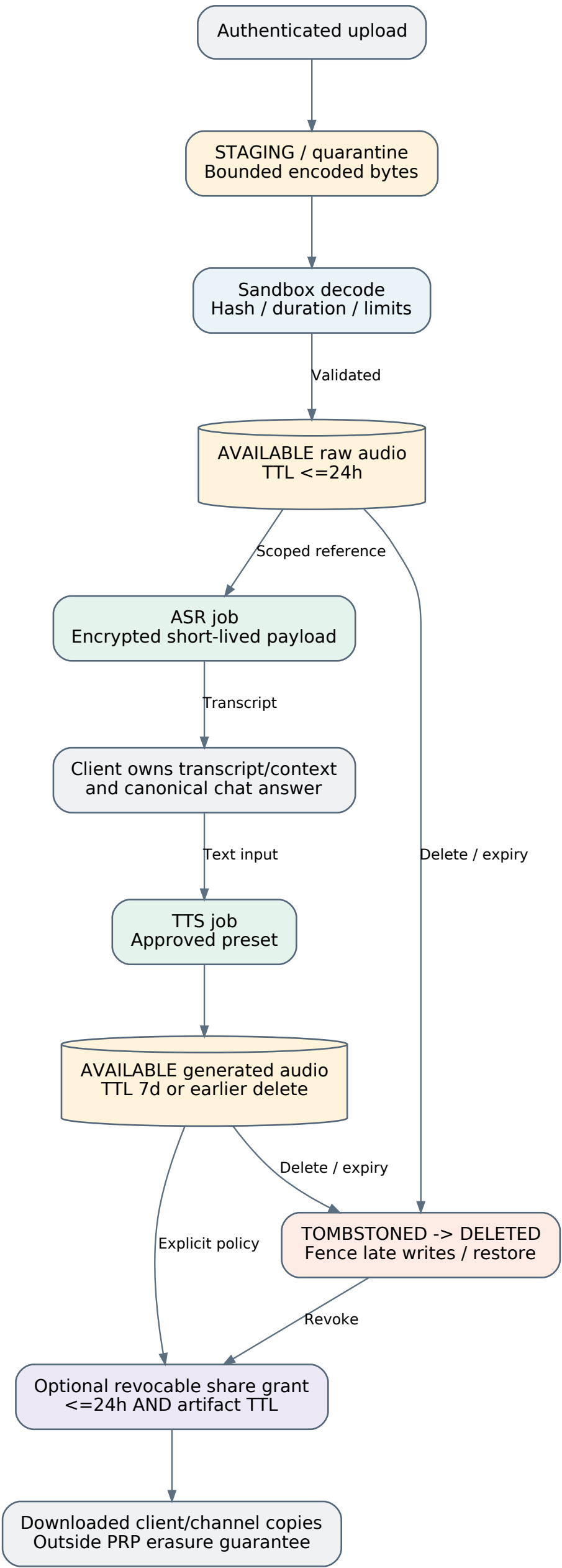


v0.3.0: logical ownership can be implemented by reused libraries/configuration; no automatic custom service.

# D19 | Audio artifact lifecycle and lineage

Data-flow + lifecycle

PRP-FR-028..030,039..041

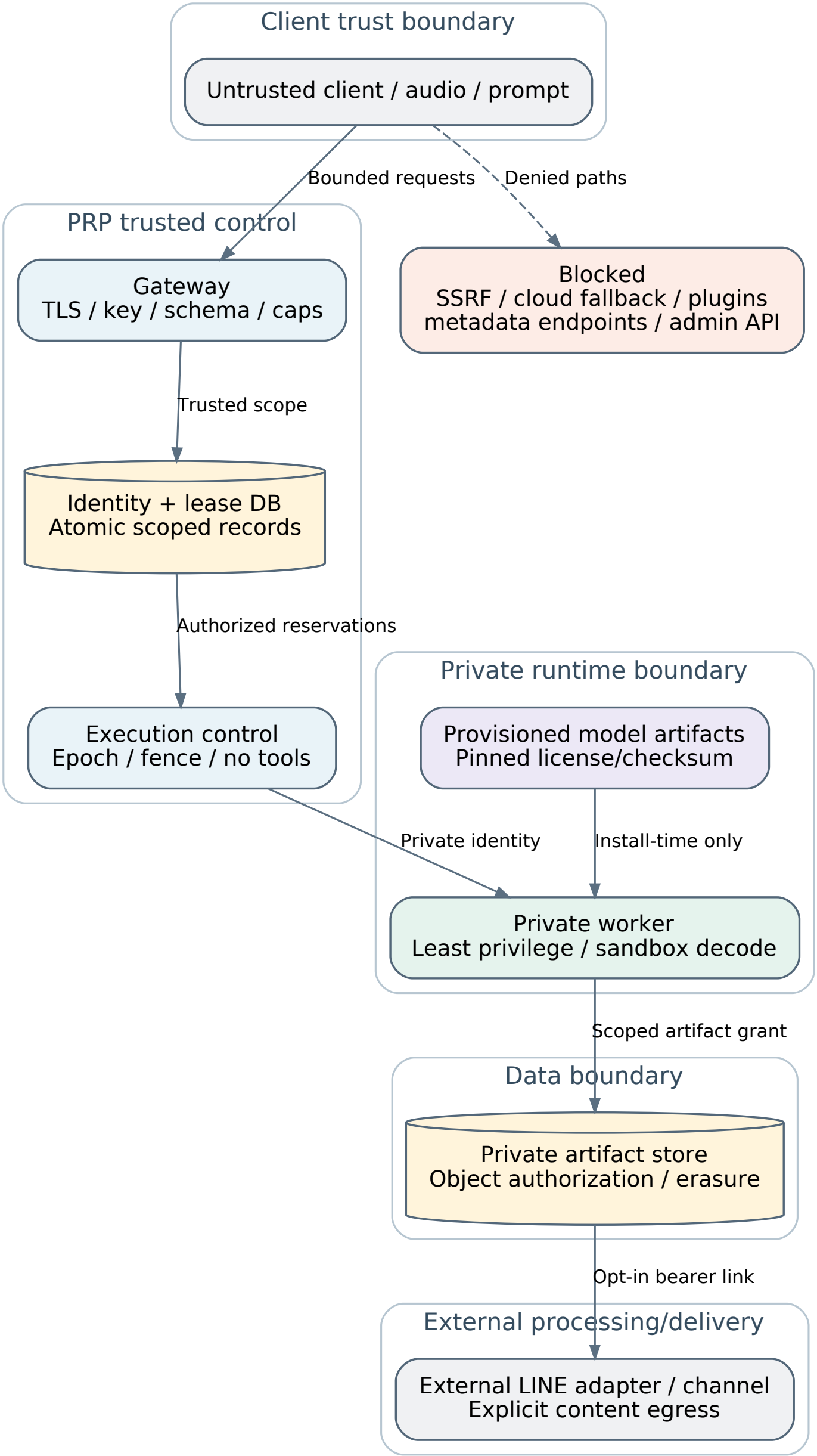


Target design. Framework bindings and hardware capacity require qualification before activation.

# D20 | Trust boundaries and threat controls

Trust-boundary / DFD security

PRP-SEC-001..012

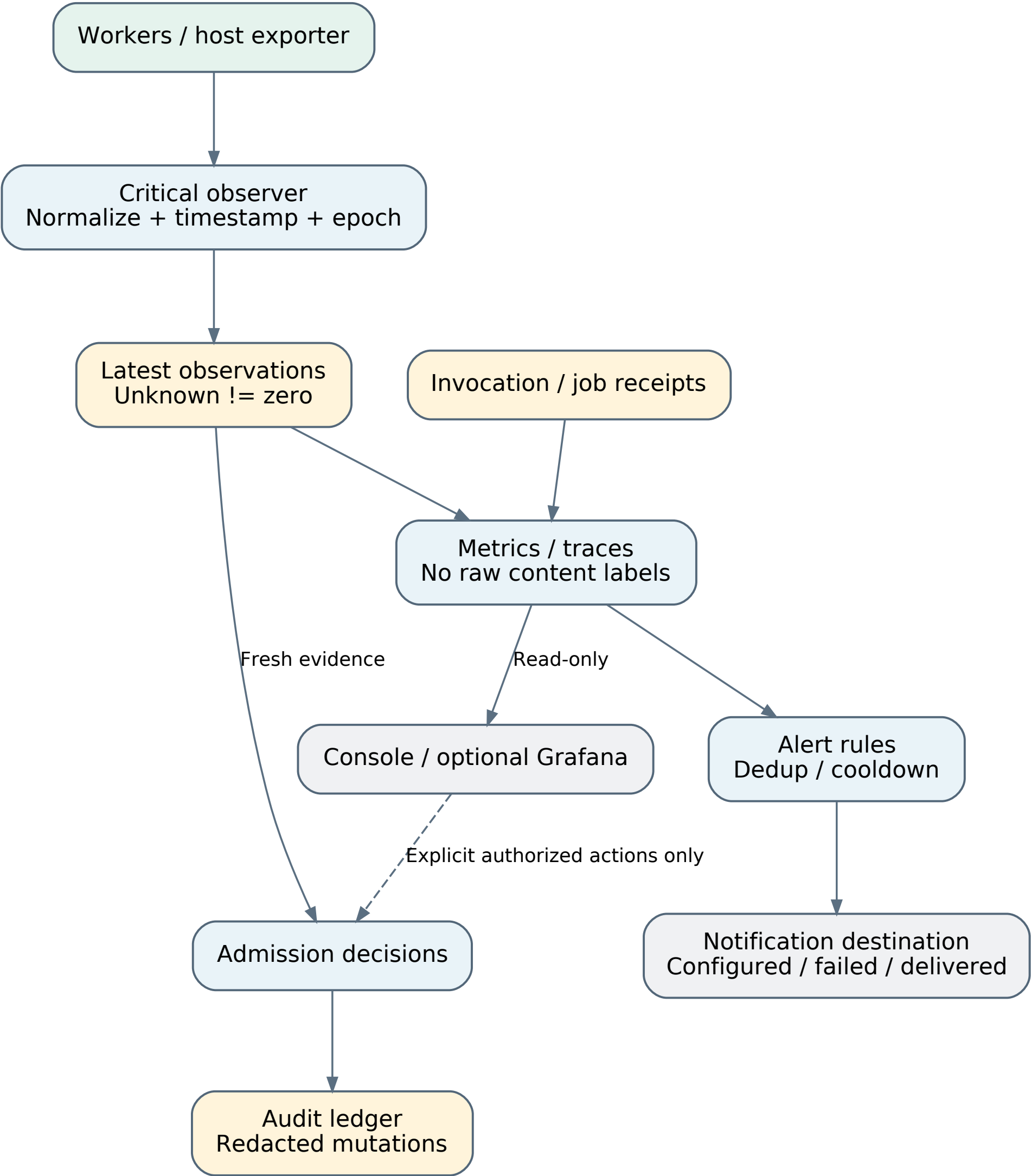


Target design. Framework bindings and hardware capacity require qualification before activation.

D21 | Observability and operational loop

Telemetry / control-flow

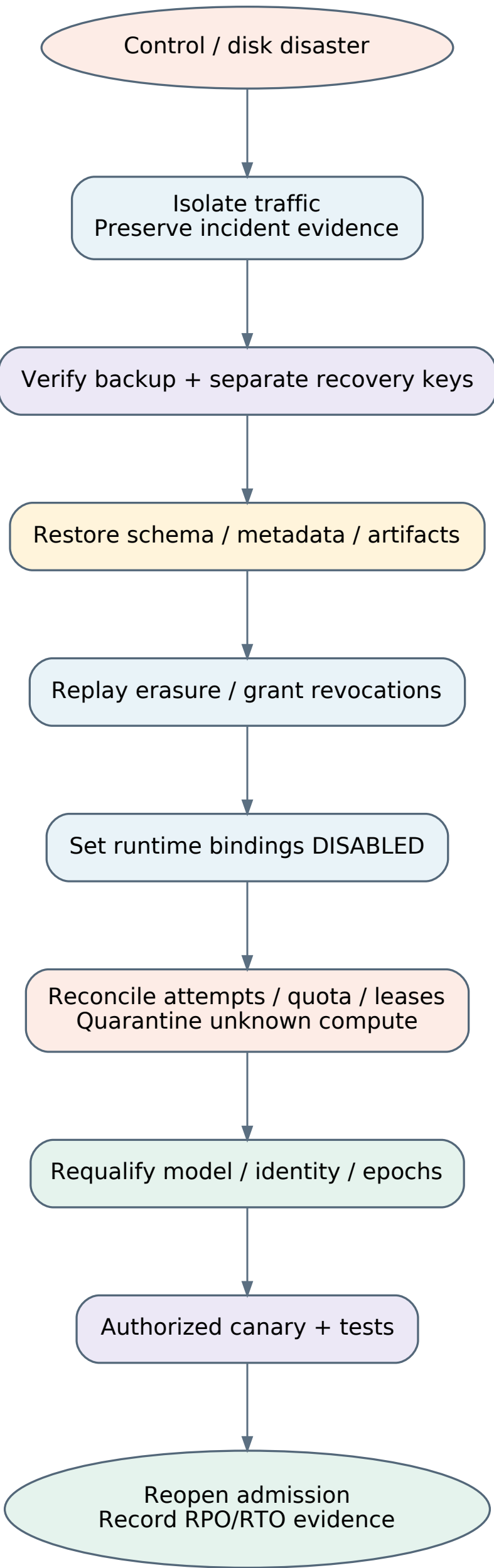
PRP-FR-014,045..047



# D22 | Disaster restore and erasure reconciliation

## Recovery activity

PRP-FR-041,051; PRP-NFR-012

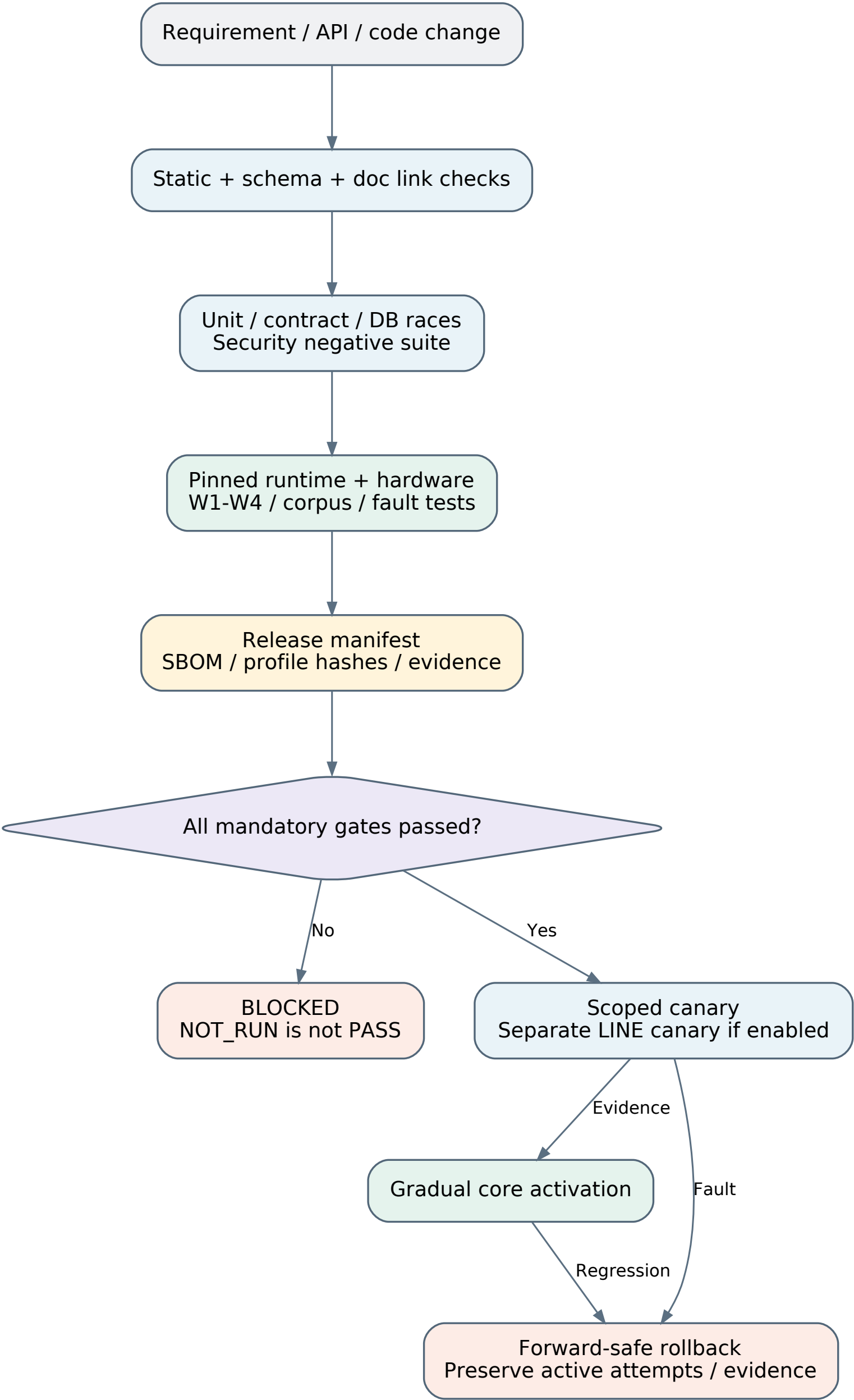


Target design. Framework bindings and hardware capacity require qualification before activation.

D23 | Release pipeline and rollback

CI/CD / release gates

PRP-FR-050..053; PRP-SEC-010

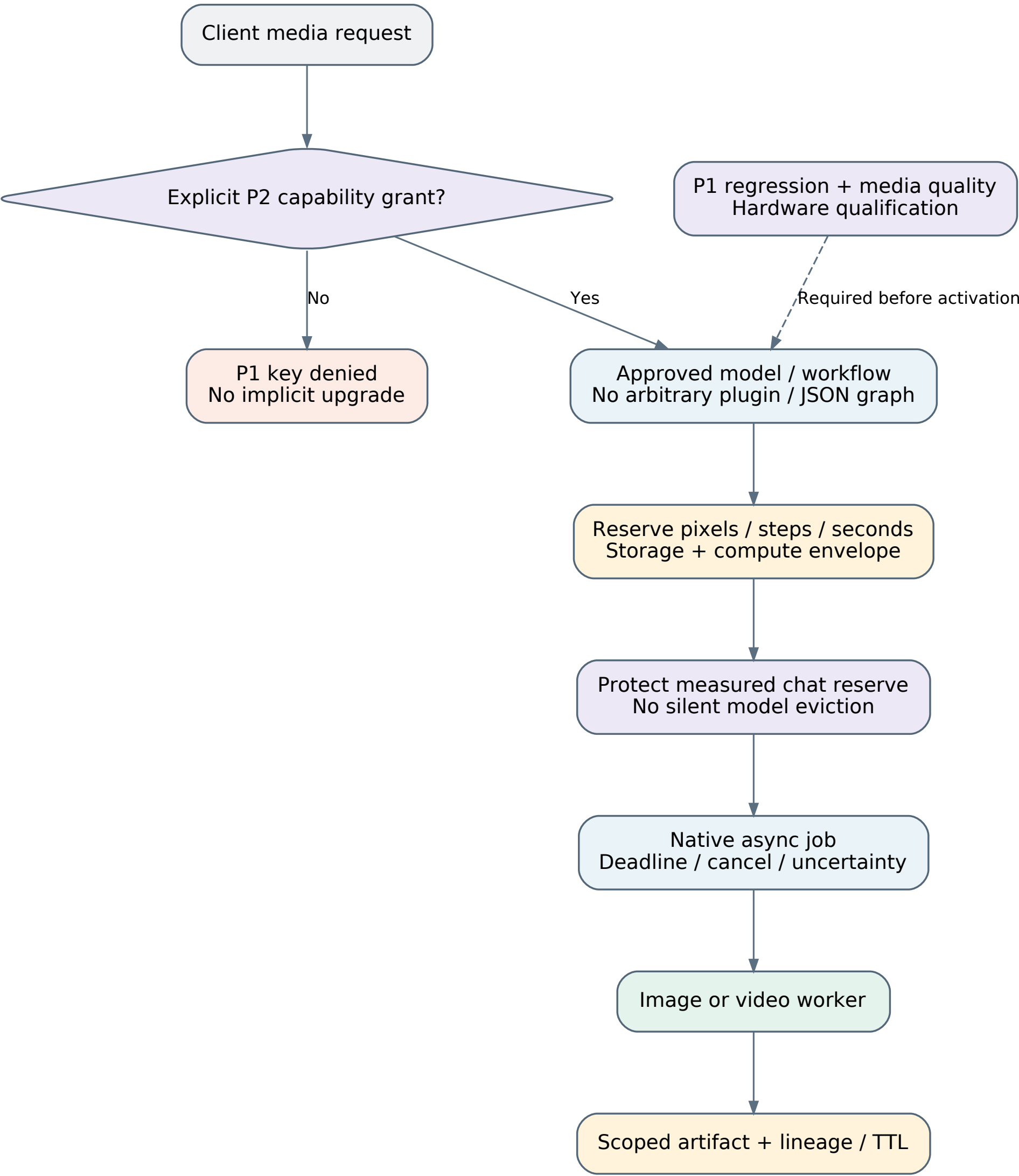


Target design. Framework bindings and hardware capacity require qualification before activation.

D24 | Phase 2 media extension

Workflow / extensibility

PRP-P2-001..008; PRP-FR-056

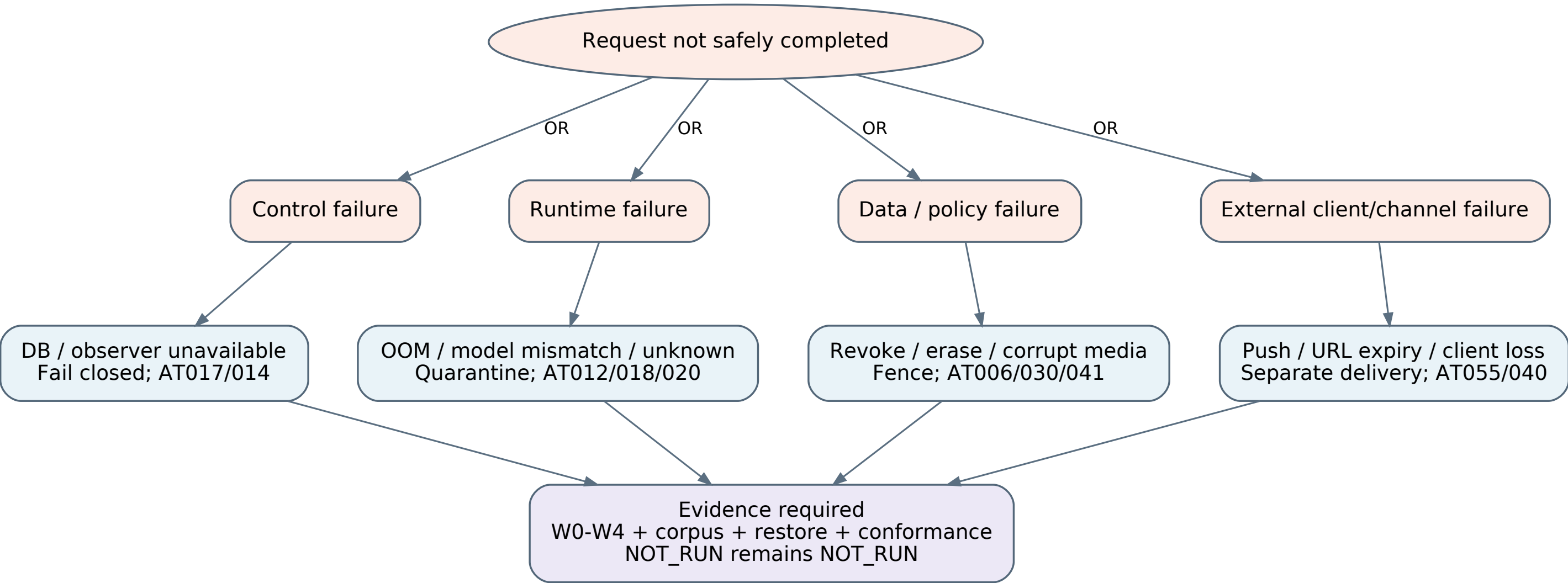


Target design. Framework bindings and hardware capacity require qualification before activation.

D25 | Failure and risk coverage

Fault tree / verification map

PRP-NFR-001..018; PRP-FR-020,051

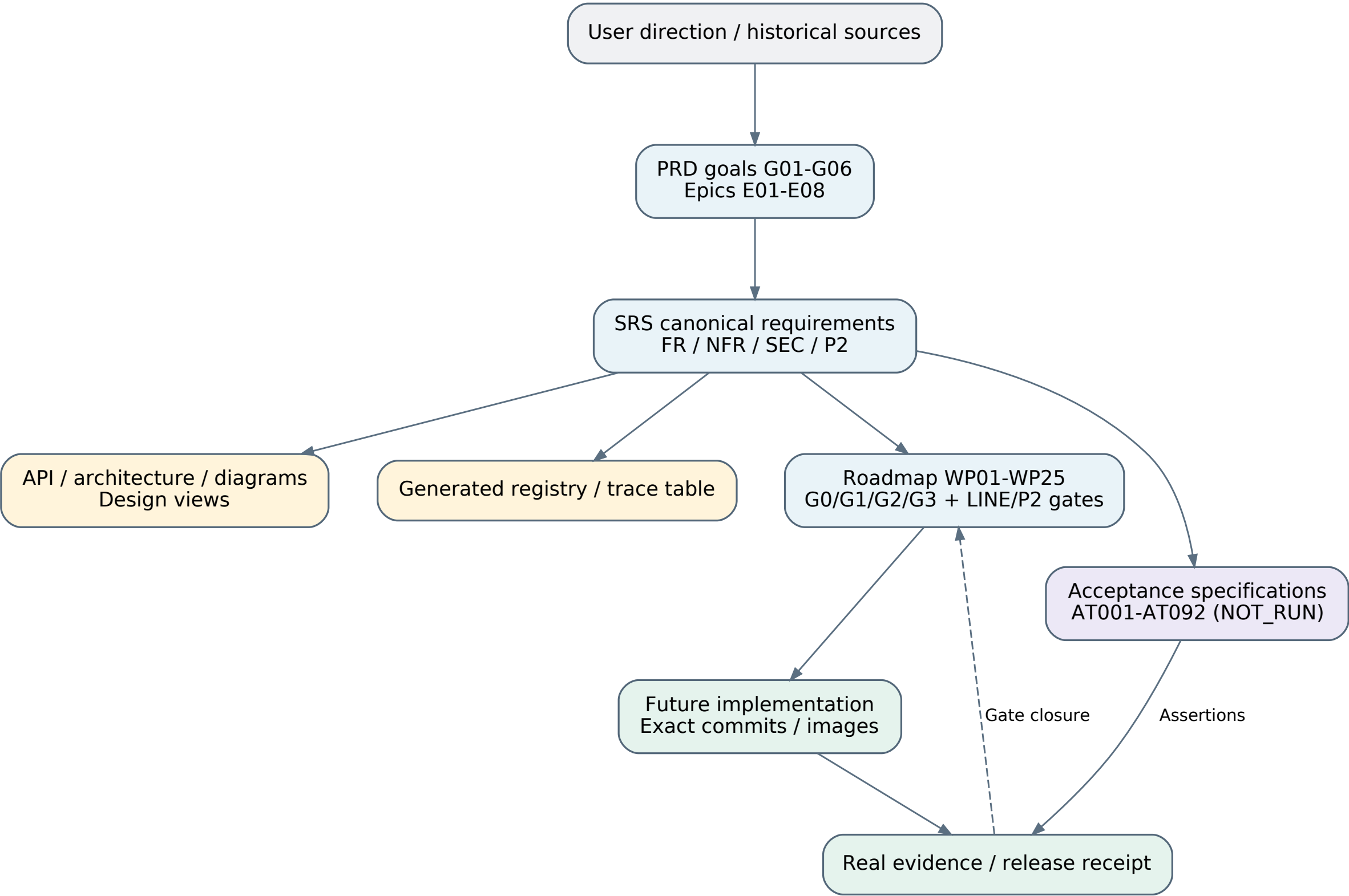


Target design. Framework bindings and hardware capacity require qualification before activation.

# D26 | Requirement-to-evidence traceability

## Traceability dependency graph

PRP-FR-053; all requirement families

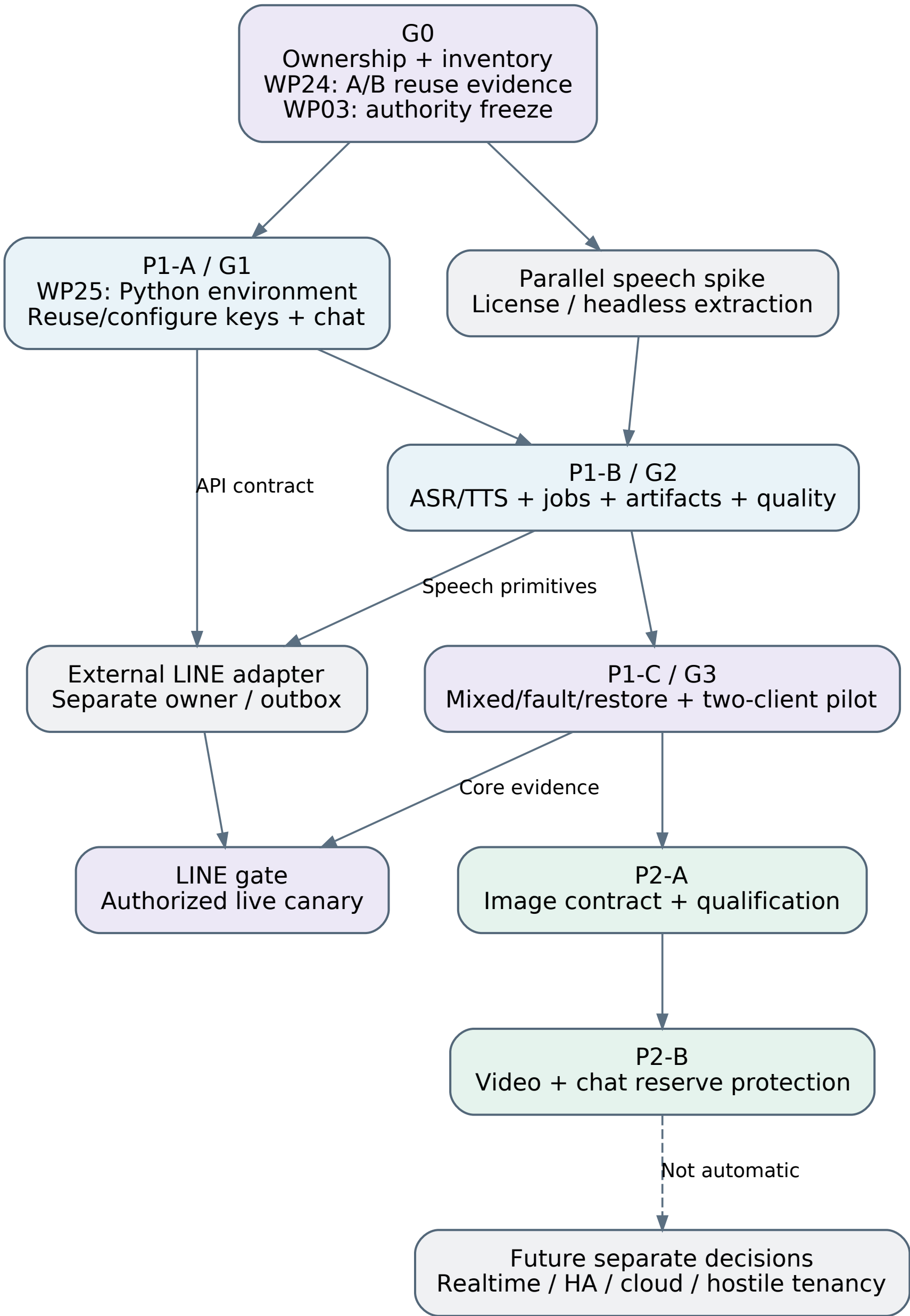


Target design. Framework bindings and hardware capacity require qualification before activation.

D27 | Roadmap dependencies and parallel lanes

Roadmap / dependency DAG

Roadmap WP01..WP25; PRP-NFR-019..024

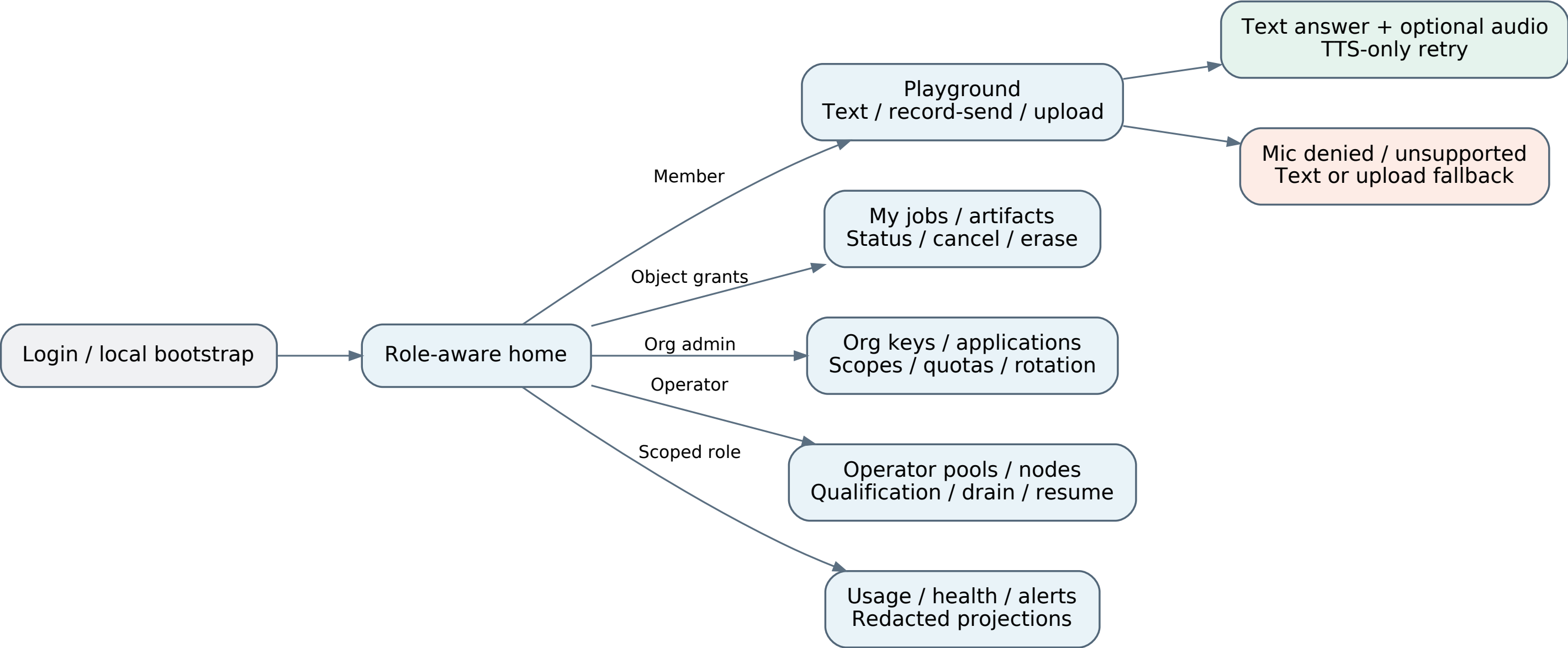


v0.3.0: logical ownership can be implemented by reused libraries/configuration; no automatic custom service.

# D28 | Console and playground navigation

UX navigation / sitemap

PRP-FR-048,049

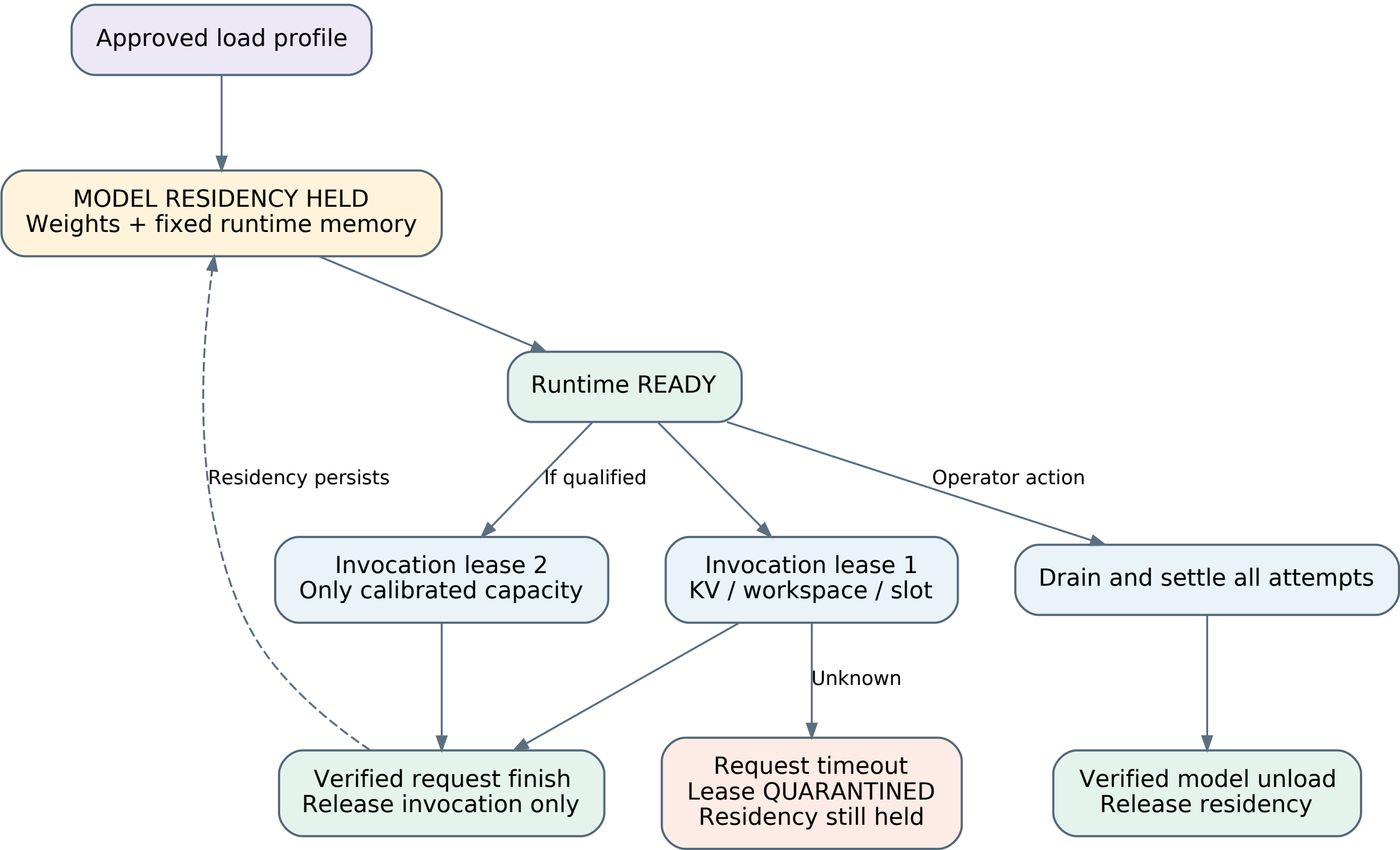


Target design. Framework bindings and hardware capacity require qualification before activation.

D29 | Resident memory versus invocation leases

Resource-allocation lifecycle

PRP-FR-013,018,044

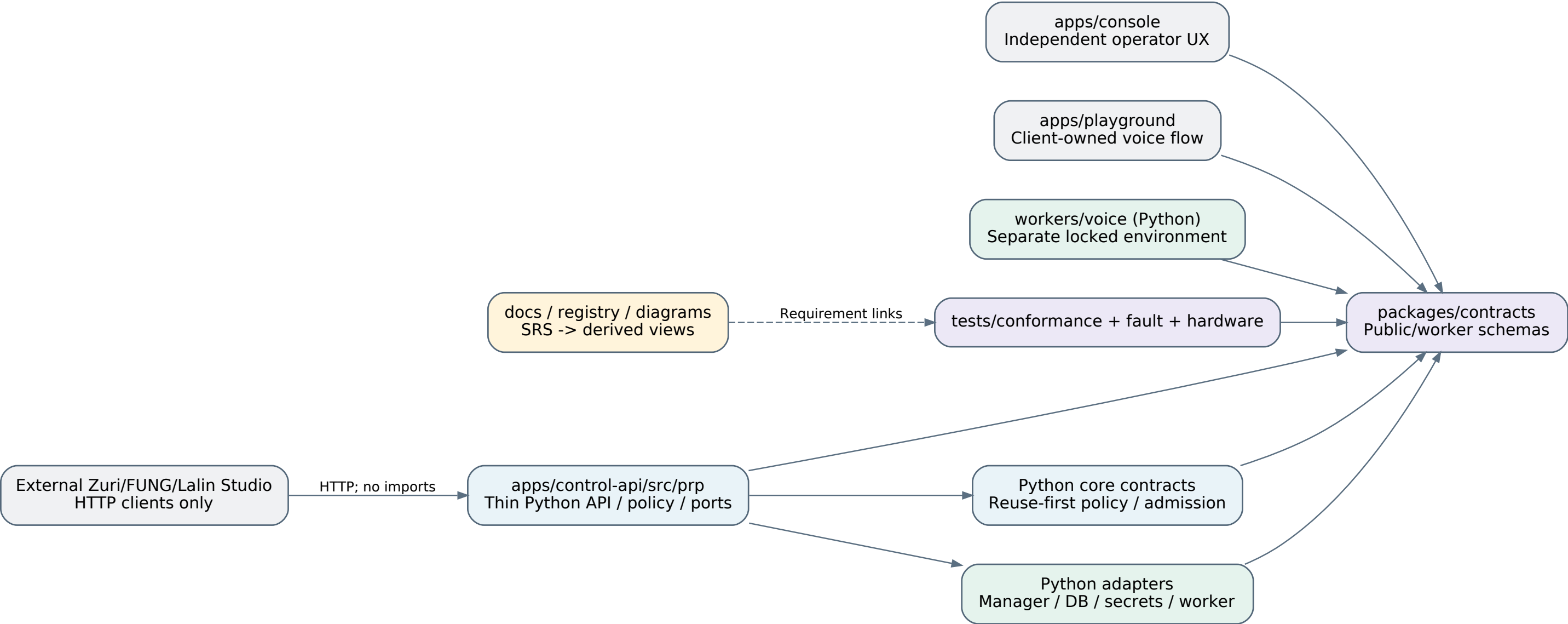


Target design. Framework bindings and hardware capacity require qualification before activation.

# D30 | Repository and package direction

## Package dependency diagram

PRP-FR-001,002,043,050,053; PRP-NFR-019,021,022

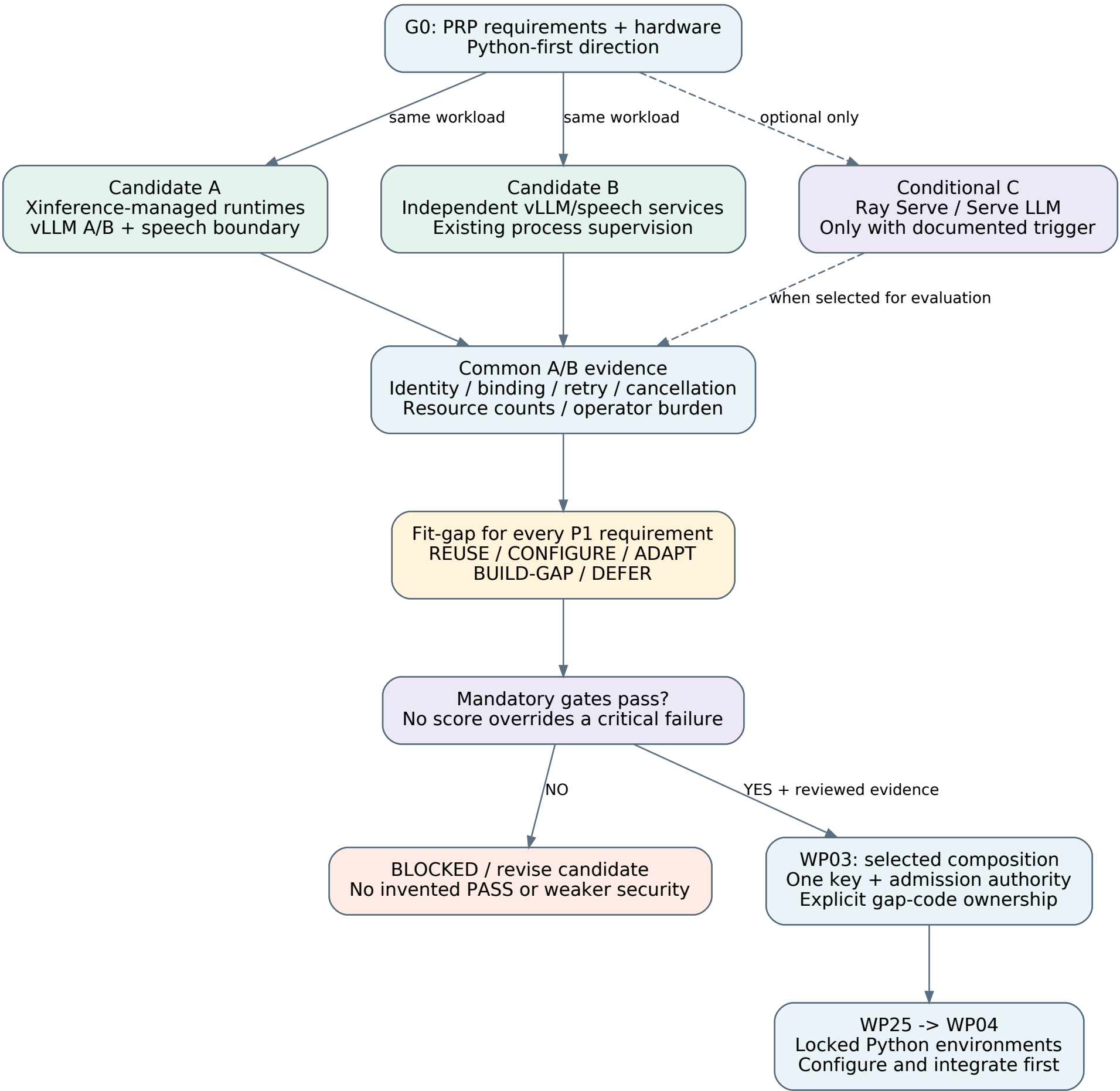


v0.3.0: logical ownership can be implemented by reused libraries/configuration; no automatic custom service.

# D31 | Reuse-first framework evaluation

Decision / evaluation DAG

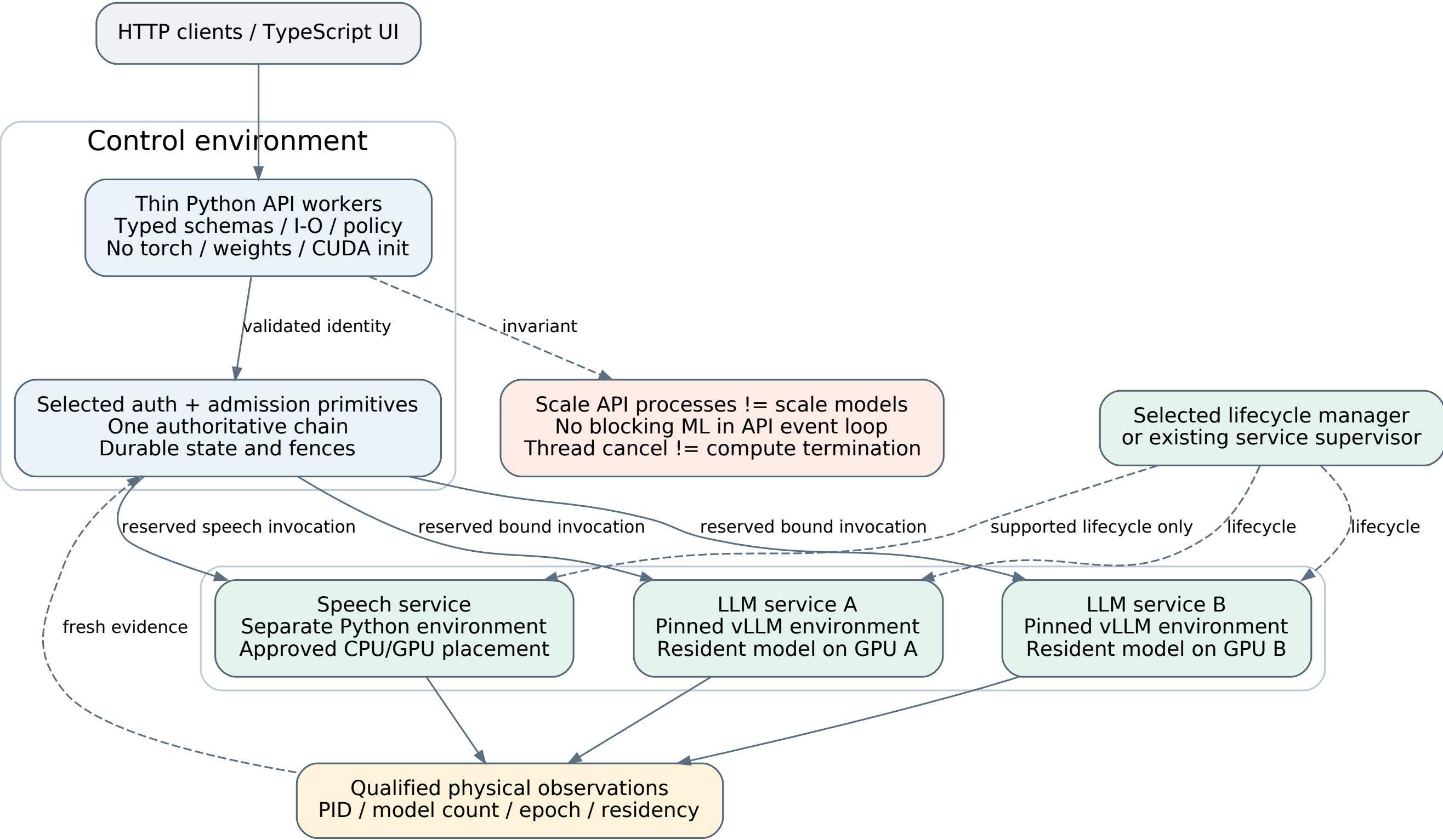
PRP-NFR-020; WP24/WP03



# D32 | Python API and model process isolation

Process / deployment boundaries

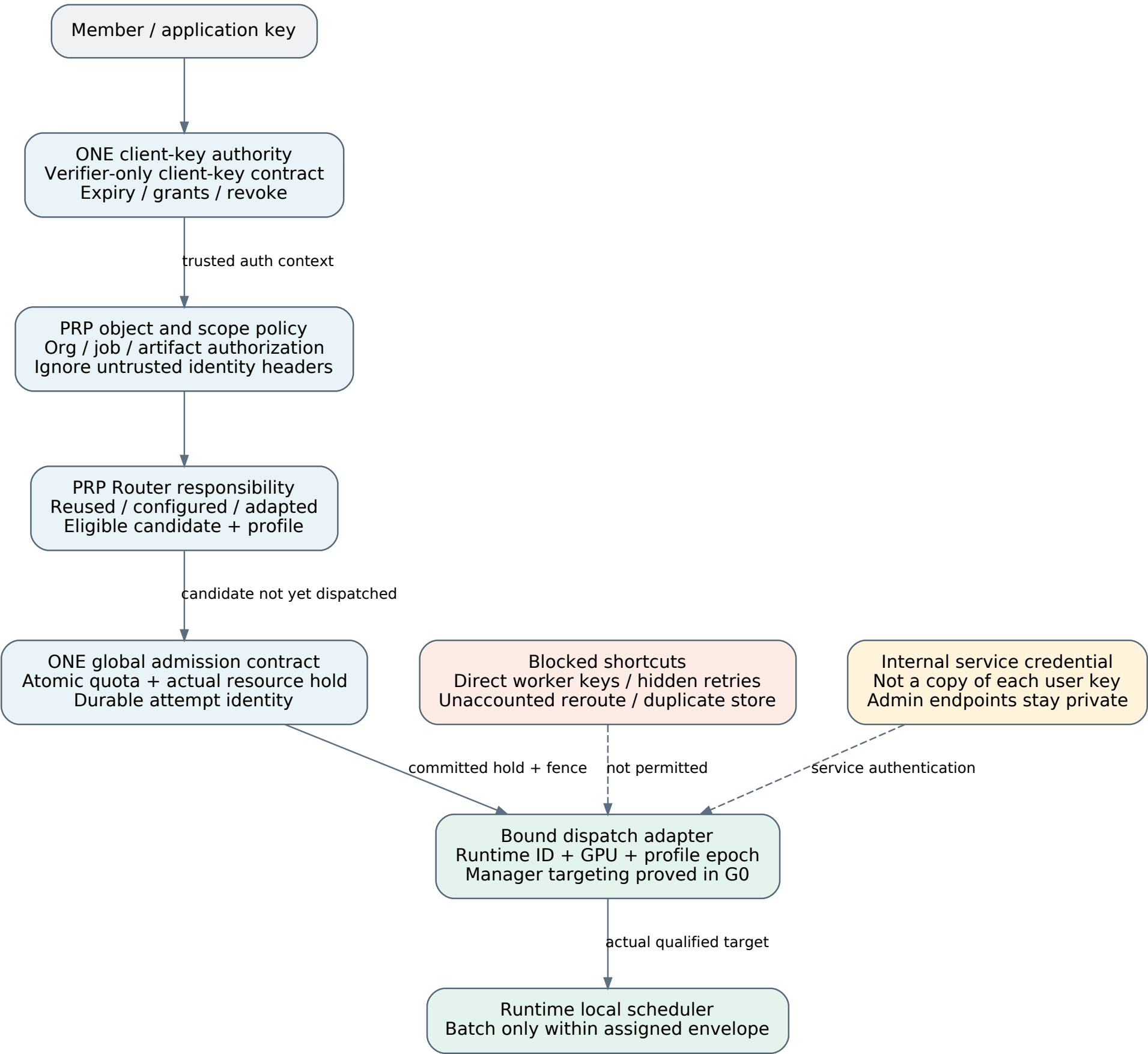
PRP-NFR-019,021,022



D33 | Framework delegation without authority bypass

Authority / admission flow

PRP-NFR-023; PRP-FR-005,017,018



# D34 | Runtime manager binding and safe lifecycle

UML-style sequence

PRP-NFR-024; PRP-FR-011,015,020,042,044

